Minimal Reality

Volume I: The Minimal Structure of Change

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Minimal Reality, Volume I

This modular monograph provides a systematic introduction to Minimal Reality as a formal mathematical structure grounded in the existence of a minimal positive change. The theory is developed through a sequence of self-contained but structurally connected papers.

Keywords: minimal reality, minimal positive change, ordered algebraic structures, discrete ordered structures, least positive element, natural-number structure.

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Paper I

The Axiom of Minimal Reality. The Minimal Threshold of Real Change

Piotr Dariusz Wójcik

Version 1.3

Abstract

This note introduces the Axiom of Minimal Reality as a minimal threshold condition for admissible magnitudes of real change. The axiom states that the ordered domain of admissible magnitudes contains a first positive element, denoted by δs. Equivalently, if

$$\mathcal{S}_{>0} := \{x \in \mathcal{S} : 0 < x\},$$

then

$$\delta s = \min \mathcal{S}_{>0}.$$

Thus there is no admissible magnitude of real change strictly between 0 and δs. The purpose of the note is not to determine the algebraic, generative, or procedural structure of magnitudes above this threshold, but only to isolate the threshold itself.

Keywords: minimal reality, real change, minimal threshold, admissible magnitude, internal discreteness.

1 Introduction

Many formal frameworks allow magnitudes of change to be arbitrarily small. In such frameworks, the passage from the absence of change to positive change may be described through ever smaller intermediate magnitudes.

The present note isolates a different formal possibility: that real change begins at a first positive admissible magnitude. This first positive magnitude is denoted by δs.

The purpose of this note is limited. It does not attempt to construct a full theory of dynamics, space, time, procedure, or dependence. It formulates only the minimal threshold condition under which a magnitude of change first becomes admissible as real.

The structure of magnitudes above this threshold is not determined here. It remains a separate question.

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2 Guiding Intuition

The guiding intuition behind the axiom is that the absence of change and the beginning of real change need not be connected by an internal continuum of ever smaller differences. There may be a first positive magnitude at which change becomes admissible as real.

This first magnitude is denoted by δs.

The axiom does not attempt to describe all structures that may arise above this threshold. It isolates only the threshold itself.

3 Preliminaries and Notation

Let $\mathcal{S}$ denote the set of admissible magnitudes of real change.

The element

$$0 \in \mathcal{S}$$

denotes the absence of change.

The relation $<$ is assumed to be a strict order on $\mathcal{S}$ and is interpreted as the order of magnitudes.

We write

$$x > 0$$

to mean

$$0 < x.$$

We also write

$$x \leq y$$

to mean that either

$$x < y$$

or

$$x = y.$$

We take 0 to be the least element of $\mathcal{S}$:

$$0 = \min \mathcal{S}.$$

Equivalently,

$$\forall x \in \mathcal{S}, \qquad 0 \leq x.$$

This is a condition on the ordered domain of admissible magnitudes. It is not part of the Axiom of Minimal Reality.

The set of positive admissible magnitudes is denoted by

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$$\mathcal{S}_{>0} := \{x \in \mathcal{S} : 0 < x\}.$$

Thus an element $x \in \mathcal{S}$ is a positive admissible magnitude of real change precisely when

$$x \in \mathcal{S}_{>0}.$$

The symbol δs will denote the first positive admissible magnitude of real change, if such a magnitude exists.

4 The Axiom

Axiom 4.1 (Minimal Reality)**.** There exists an element $\delta s \in \mathcal{S}_{>0}$ such that for every $x \in \mathcal{S}_{>0}$,

$$\delta s \leq x.$$

The element δs is called the *minimal real magnitude of change*.

Equivalently, δs is the least positive element of $\mathcal{S}$:

$$\delta s = \min \mathcal{S}_{>0}.$$

The axiom does not assign any metric value to δs. It states only that δs is the first positive magnitude admitted by the ordered structure $\mathcal{S}$.

5 Immediate Consequences

Proposition 5.1 (No intermediate admissible magnitude)**.** *There is no admissible magnitude of real change strictly between* 0 *and* δs*.*

Proof. Suppose that there exists $x \in \mathcal{S}$ such that

$$0 < x < \delta s.$$

Since $0 < x$, we have

$$x \in \mathcal{S}_{>0}.$$

By the Axiom of Minimal Reality,

$$\delta s \leq x.$$

This contradicts

$$x < \delta s.$$

Therefore no such x exists. □

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Hence

$$\neg\exists x \in \mathcal{S} \quad (0 < x < \delta s).$$

Moreover, by the Axiom of Minimal Reality, every positive admissible magnitude of real change is at least δs:

$$\forall x \in \mathcal{S} \quad (x > 0 \Longrightarrow \delta s \leq x).$$

6 Threshold Discreteness

The Axiom of Minimal Reality excludes the existence of admissible positive magnitudes of real change below δs. Internally, this means

$$\neg\exists \epsilon \in \mathcal{S} \quad (0 < \epsilon < \delta s).$$

If ϵ is represented in an external ordered language, then

$$0 < \epsilon < \delta s \quad \Longrightarrow \quad \epsilon \notin \mathcal{S}.$$

Thus an expression may be formally representable in an external language without being an admissible magnitude of real change inside $\mathcal{S}$.

In particular, limiting expressions of the form

$$h \to 0$$

do not describe an internal descent through positive admissible magnitudes of $\mathcal{S}$ below δs.

This does not prevent the use of continuous mathematics as an external language of approximation or description. The axiom states here only that no positive admissible magnitude lies strictly between 0 and δs.

7 Scope of the Axiom

The Axiom of Minimal Reality establishes the existence of a minimal positive admissible magnitude of real change. It does not yet determine the full algebraic, generative, or procedural structure of magnitudes above this threshold.

In particular, the axiom does not determine how magnitudes above δs are composed, generated, compared by additional operations, or organized into higher structures.

It also does not determine procedures, trajectories, orientations, operators, variables, dependencies, probabilities, space, or time.

The structural organization of magnitudes above the minimal threshold is left for subsequent work.

Thus the axiom states only the following:

$$\boxed{\delta s = \min \mathcal{S}_{>0}.}$$

Consequently,

$$\boxed{\neg\exists x \in \mathcal{S} \quad (0 < x < \delta s).}$$

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8 Conclusion

The Axiom of Minimal Reality isolates the minimal threshold at which a magnitude of change becomes admissible as real. It states that the ordered domain of admissible magnitudes contains a first positive element, denoted by δs.

The essential content of the axiom is

$$\delta s = \min \mathcal{S}_{>0},$$

and therefore

$$\neg \exists x \in \mathcal{S} \quad (0 < x < \delta s).$$

The axiom isolates the threshold itself. The structure above the threshold remains a separate question.

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Paper II

The Structure of Real Magnitudes of Change. Finite Generation Above the Minimal Threshold

Piotr Dariusz Wójcik

Version 1.3

Abstract

This paper studies the structural organization of admissible magnitudes of real change above the minimal threshold introduced in the Axiom of Minimal Reality. The threshold axiom establishes a first positive admissible magnitude,

$$\delta s = \min \mathcal{S}_{>0}, \qquad \mathcal{S}_{>0} := \{x \in \mathcal{S} : 0 < x\}.$$

It does not, by itself, determine the structure of magnitudes above this threshold. The present paper considers the finitely generated structural model, in which every admissible magnitude is a finite composition of δs. Under the finite generation principle, the structure of admissible magnitudes has the form

$$\mathcal{S} = \{0, \delta s, 2\delta s, 3\delta s, \dots\}.$$

In the finitely generated model considered here, the order already present on $(\mathcal{S})$ is additionally required to agree with the number of minimal compositions:

$$n\delta s < m\delta s \quad \Longleftrightarrow \quad n < m.$$

With this compatibility condition, the structure is represented by the ordered commutative monoid

$$(\mathbb{N}_0, +, 0, <).$$

Finite generation is not introduced as a second axiom of Minimal Reality, but as a structural principle selecting one natural model of admissible magnitudes above the minimal threshold.

Keywords: minimal reality, real magnitudes, finite generation, minimal threshold, ordered monoid, structural model.

1 Introduction

The Axiom of Minimal Reality establishes the existence of a first positive admissible magnitude of real change. If $\mathcal{S}$ denotes the ordered domain of admissible magnitudes and

$$\mathcal{S}_{>0} := \{x \in \mathcal{S} : 0 < x\},$$

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then the axiom states that there exists an element δs such that

$$\delta s = \min \mathcal{S}_{>0}.$$

This axiom determines the threshold at which real change begins. It says that no admissible magnitude of real change lies strictly between 0 and δs:

$$\neg \exists x \in \mathcal{S} \quad (0 < x < \delta s).$$

However, the threshold axiom does not determine how magnitudes above δs are organized. It does not say whether every admissible magnitude above the threshold is built from δs, whether admissible magnitudes not generated by finite compositions of δs are allowed, or whether the structure above the threshold is generated, partially generated, or enriched by further elements.

The purpose of this paper is to study the structural organization of admissible magnitudes above the minimal threshold.

The structural model studied here is the finitely generated model. In this model, every admissible magnitude of real change is a finite composition of δs:

$$\forall x \in \mathcal{S} \quad \exists! n \in \mathbb{N}_0 \quad x = n\delta s.$$

This condition will be called the finite generation principle. It is not a second axiom of Minimal Reality. It is an additional structural principle describing one distinguished organization of magnitudes above the minimal threshold.

2 Why This Second Step Is Needed

The threshold axiom gives the first positive admissible magnitude:

$$\delta s = \min \mathcal{S}_{>0}.$$

This is enough to establish the absence of admissible magnitudes strictly between 0 and δs. It is not enough to determine the entire structure of $\mathcal{S}$ above δs.

In particular, from

$$\delta s = \min \mathcal{S}_{>0}$$

it does not follow that

$$\mathcal{S} = \{0, \delta s, 2\delta s, 3\delta s, \dots\}.$$

The implication

$$\delta s = \min \mathcal{S}_{>0} \quad \Longrightarrow \quad \mathcal{S} = \{n\delta s : n \in \mathbb{N}_0\}$$

is not valid without an additional structural condition.

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Indeed, it is possible to have a structure in which δs is the least positive admissible magnitude, while some element $y \in \mathcal{S}$ satisfies

$$y > \delta s$$

and yet is not a finite composition of δs:

$$y \neq n\delta s \qquad \text{for every } n \in \mathbb{N}_0.$$

Such a structure satisfies the threshold axiom but does not satisfy finite generation.

Therefore a second step is needed if one wants to pass from the existence of a minimal threshold to a generated structure of all admissible magnitudes above that threshold. This second step is not a new foundation of Minimal Reality. It is a structural study of what may be built above the foundation already given by the threshold axiom.

3 From Threshold to Structural Models

The transition from the threshold axiom to a generated structure is not a logical consequence of the threshold alone. It is a structural choice.

The ordered domain inherited from Paper I satisfies

$$0 = \min \mathcal{S}.$$

This is a condition on the domain and is not part of the threshold axiom.

On this domain, the threshold axiom establishes:

$$\delta s = \min \mathcal{S}_{>0}.$$

A generated structural model adds the claim that every admissible magnitude is obtained by finite repetition of this minimal magnitude:

$$\forall x \in \mathcal{S} \quad \exists! n \in \mathbb{N}_0 \quad x = n\delta s.$$

These are distinct levels.

The first level is the level of the threshold:

there exists a first positive admissible magnitude.

The second level is the level of structural organization:

all admissible magnitudes are generated from that magnitude.

The purpose of this paper is to study the second level while preserving the distinction between the two.

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4 Algebraic Preliminaries

In order to speak about finite generation, we extend the threshold structure by adding an operation of composition of magnitudes.

Let

$$(\mathcal{S}, +, 0, <)$$

be a structure where $\mathcal{S}$ is the set of admissible magnitudes of real change, $0 \in \mathcal{S}$ denotes the absence of change, $<$ denotes the order of magnitudes, and

$$+ : \mathcal{S} \times \mathcal{S} \to \mathcal{S}$$

denotes the composition of magnitudes.

At the level of magnitudes, we assume that $+$ is associative and commutative, with 0 as its neutral element:

$$(x + y) + z = x + (y + z),$$

$$x + y = y + x,$$

and

$$x + 0 = x$$

for all $x, y, z \in \mathcal{S}$.

Thus

$$(\mathcal{S}, +, 0)$$

is a commutative monoid.

The commutativity of $+$ belongs only to the level of magnitudes. It expresses that $\mathcal{S}$ records how much change has occurred, not the procedural order in which change has been performed. No claim about procedural order is made in this paper.

We also use

$$\mathbb{N}_0 = \{0, 1, 2, 3, \dots\}$$

for the set of natural numbers including zero.

5 The Minimal Unit and Repeated Composition

Let δs be the minimal real magnitude of change established by the threshold axiom:

$$\delta s = \min \mathcal{S}_{>0}.$$

For $n \in \mathbb{N}_0$, the expression

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$$n\delta s$$

denotes the n-fold composition of δs.

It is defined recursively by

$$0 \cdot \delta s := 0,$$

and

$$(n+1)\delta s := n\delta s + \delta s.$$

Thus,

$$1\delta s = \delta s,$$

$$2\delta s = \delta s + \delta s,$$

$$3\delta s = (\delta s + \delta s) + \delta s,$$

and so on.

The notation $n\delta s$ should not be read as scalar multiplication in a vector space. It is notation for repeated composition in the commutative monoid

$$(\mathcal{S}, +, 0).$$

6 The Status of Finite Generation

The finite generation principle is not introduced as a second axiom of Minimal Reality.

The threshold axiom alone states:

$$\delta s = \min \mathcal{S}_{>0}.$$

It establishes the first positive admissible magnitude of real change. It does not determine the full algebraic or generative structure above that threshold.

Finite generation belongs to a different level. It is a structural principle selecting one natural model of admissible magnitudes above the minimal threshold.

Thus the relation between the two levels is:

$$\text{Minimal Reality} \quad = \quad \text{threshold axiom},$$

whereas

$$\text{finite generation} \quad = \quad \text{structural organization above the threshold.}$$

The present paper studies this structural organization. It does not alter the content of the original threshold axiom.

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7 Motivation for Finite Generation

The finite generation principle is not forced by the threshold axiom alone. It is an additional structural condition selecting the model in which every admissible magnitude is generated by finite composition of the minimal magnitude δs:

$$\forall x \in \mathcal{S} \quad \exists! n \in \mathbb{N}_0 \quad x = n\delta s.$$

The threshold axiom itself establishes only

$$\delta s = \min \mathcal{S}_{>0},$$

and hence

$$\neg \exists x \in \mathcal{S} \quad (0 < x < \delta s).$$

It does not determine how all admissible magnitudes above the threshold are generated.

The finitely generated model adds the requirement that every admissible magnitude belongs to the generated family

$$\{n\delta s : n \in \mathbb{N}_0\}.$$

Consequently,

$$\mathcal{S} = \{0, \delta s, 2\delta s, 3\delta s, \dots\}.$$

Thus finite generation expresses the precise structural condition that no admissible magnitudes occur outside the finite compositions of δs.

8 Alternative Models Above the Threshold

The finite generation model is not the only logically possible model above the minimal threshold. It is useful to distinguish it from other possibilities.

8.1 Threshold-Only Models

A threshold-only model satisfies

$$\delta s = \min \mathcal{S}_{>0},$$

but does not require every admissible magnitude to be a finite composition of δs.

Such a model may contain an element $y \in \mathcal{S}$ such that

$$y > \delta s,$$

while

$$y \neq n\delta s \qquad \text{for every } n \in \mathbb{N}_0.$$

This model preserves the minimal threshold but does not impose finite generation.

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8.2 Finitely Generated Models

A finitely generated model satisfies

$$\forall x \in \mathcal{S} \quad \exists! n \in \mathbb{N}_0 \quad x = n\delta s.$$

In such a model,

$$\mathcal{S} = \{0, \delta s, 2\delta s, 3\delta s, \dots\}.$$

This is the model studied in the present paper.

8.3 Partially Generated or Enriched Models

A partially generated or enriched model may contain all finite compositions of δs, together with additional admissible magnitudes above the threshold:

$$\mathcal{S} = \{0, \delta s, 2\delta s, 3\delta s, \dots\} \cup E,$$

where E is a set of additional elements satisfying

$$e > \delta s \qquad \text{for all } e \in E.$$

For such elements, it need not be true that

$$e = n\delta s$$

for some $n \in \mathbb{N}_0$.

Such a model still may satisfy the minimal threshold axiom, but it does not satisfy full finite generation.

9 Comparison with Non-Generative Models

Non-generative and enriched models are logically possible. Their possibility is important, because it shows that finite generation is not a consequence of the threshold axiom alone.

If a model contains an element $y \in \mathcal{S}$ such that

$$y > \delta s$$

and

$$y \neq n\delta s \qquad \text{for every } n \in \mathbb{N}_0,$$

then y is an admissible magnitude not generated by finite composition of δs.

Similarly, in an enriched model

$$\mathcal{S} = \{0, \delta s, 2\delta s, 3\delta s, \dots\} \cup E,$$

each element

$$e \in E$$

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with

$$e \neq n\delta s \qquad \text{for every } n \in \mathbb{N}_0$$

is likewise an admissible magnitude outside the family generated by δs.

Such models preserve the minimal threshold but contain admissible magnitudes outside the family generated by finite compositions of δs.

The finitely generated model is distinguished by the stronger condition

$$\forall x \in \mathcal{S} \quad \exists! n \in \mathbb{N}_0 \quad x = n\delta s.$$

Thus every admissible magnitude in that model is generated from δs by finite composition.

10 The Finite Generation Principle

We now state the central structural principle considered in this paper.

Principle 10.1 (Finite Generation)**.** *Every admissible magnitude of real change is uniquely expressible as a finite composition of the minimal real magnitude δs:*

$$\forall x \in \mathcal{S} \quad \exists! n \in \mathbb{N}_0 \quad x = n\delta s.$$

The Finite Generation Principle is not introduced as a second axiom of Minimal Reality. It is introduced as a structural condition on the organization of admissible magnitudes above the minimal threshold.

It says that the threshold is not only the first positive admissible magnitude, but also the generator of all admissible magnitudes in the model under consideration.

11 The Generated Structure

Theorem 11.1 (Generated Form of $\mathcal{S}$)**.** *Under the Finite Generation Principle,*

$$\mathcal{S} = \{n\delta s : n \in \mathbb{N}_0\}.$$

Equivalently,

$$\mathcal{S} = \{0, \delta s, 2\delta s, 3\delta s, \dots\}.$$

Proof. By the Finite Generation Principle, for every $x \in \mathcal{S}$ there exists a unique $n \in \mathbb{N}_0$ such that

$$x = n\delta s.$$

Therefore,

$$\mathcal{S} \subseteq \{n\delta s : n \in \mathbb{N}_0\}.$$

Conversely, by the recursive definition of repeated composition,

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$$0 \cdot \delta s = 0 \in \mathcal{S},$$

and if

$$n\delta s \in \mathcal{S},$$

then

$$(n+1)\delta s = n\delta s + \delta s \in \mathcal{S}$$

because $+$ is an operation on $\mathcal{S}$. Hence

$$n\delta s \in \mathcal{S}$$

for every $n \in \mathbb{N}_0$. Thus,

$$\{n\delta s : n \in \mathbb{N}_0\} \subseteq \mathcal{S}.$$

Combining the two inclusions gives

$$\mathcal{S} = \{n\delta s : n \in \mathbb{N}_0\}.$$

□

12 Ordered Monoid Representation

The finitely generated structure admits a natural ordered monoid representation.

In order to obtain an ordered representation, finite generation must be distinguished from compatibility of the order with the generated levels. In the finitely generated model considered here, we require the order already present on $(\mathcal{S})$ to satisfy

$$n\delta s < m\delta s \quad \Longleftrightarrow \quad n < m \qquad (n, m \in \mathbb{N}_0).$$

This is a compatibility condition of the finitely generated structural model. It is not a consequence of the Finite Generation Principle alone and does not modify the threshold axiom.

Proposition 12.1 (Ordered monoid representation)**.** *Under the Finite Generation Principle and the generation-compatible order condition,*

$$(\mathcal{S}, +, 0, <) \cong (\mathbb{N}_0, +, 0, <).$$

Proof. Define

$$\varphi : \mathbb{N}_0 \to \mathcal{S}$$

by

$$\varphi(n) = n\delta s.$$

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By the Finite Generation Principle, φ is bijective. Moreover,

$$\varphi(0) = 0 \cdot \delta s = 0.$$

For $n, m \in \mathbb{N}_0$,

$$\varphi(n + m) = (n + m)\delta s.$$

By the recursive definition of repeated composition,

$$(n + m)\delta s = n\delta s + m\delta s.$$

Thus

$$\varphi(n + m) = \varphi(n) + \varphi(m).$$

Hence φ is an isomorphism of commutative monoids:

$$(\mathbb{N}_0, +, 0) \cong (\mathcal{S}, +, 0).$$

By the generation-compatible order condition,

$$n\delta s < m\delta s \quad \Longleftrightarrow \quad n < m.$$

With this order,

$$\varphi(n) < \varphi(m) \quad \Longleftrightarrow \quad n < m.$$

Therefore φ is an isomorphism of ordered commutative monoids:

$$(\mathcal{S}, +, 0, <) \cong (\mathbb{N}_0, +, 0, <).$$

□

Remark 12.2 (Exact Structural Status). The isomorphism

$$(\mathcal{S}, +, 0, <) \cong (\mathbb{N}_0, +, 0, <)$$

is exact at the level of ordered commutative monoids. In particular, δs corresponds to (1), and every admissible magnitude in the finitely generated model corresponds uniquely to a natural-number level.

No additional algebraic distinction from $(\mathbb{N}_0, +, 0, <)$ is claimed at this level. The terminology of admissible magnitudes and minimal real change records the interpretation and genealogy assigned to the structure in the present theory, not an additional ordered-monoid structure absent from the classical model.

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13 Relation to the Threshold Axiom

The threshold axiom states:

$$\delta s = \min \mathcal{S}_{>0}.$$

The finite generation model adds:

$$\forall x \in \mathcal{S} \quad \exists! n \in \mathbb{N}_0 \quad x = n\delta s.$$

In the finitely generated model considered here, the inherited order is additionally required to be generation-compatible:

$$n\delta s < m\delta s \quad \Longleftrightarrow \quad n < m.$$

Under finite generation together with the generation-compatible order condition, the generated model satisfies the threshold condition.

Indeed, the positive elements of $\mathcal{S}$ have the form

$$n\delta s$$

with

$$n \geq 1.$$

The least such element is obtained when

$$n = 1.$$

Hence,

$$\delta s = \min \mathcal{S}_{>0}.$$

Thus,

$$\boxed{\text{Finite Generation} + \text{generation-compatible order} \quad \Longrightarrow \quad \text{Minimal Threshold.}}$$

The converse implication does not hold:

$$\text{Minimal Threshold} \quad \not\Longrightarrow \quad \text{Finite Generation.}$$

The existence of a minimal positive admissible magnitude does not by itself imply that every admissible magnitude is a finite composition of that magnitude.

Therefore the finitely generated model is stronger than the threshold axiom. It is a structural development of it, not a logical consequence of it alone.

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14 Internal Discreteness Revisited

The threshold axiom gives the first form of internal discreteness:

$$\neg\exists x \in \mathcal{S} \quad (0 < x < \delta s).$$

The finitely generated model gives a stronger structural form:

$$\mathcal{S} = \{0, \delta s, 2\delta s, 3\delta s, \dots\}.$$

Thus the admissible magnitudes form a sequence of finite levels:

$$0 \longrightarrow \delta s \longrightarrow 2\delta s \longrightarrow 3\delta s \longrightarrow \cdots .$$

This sequence should not be interpreted as a procedural trajectory. It is an ordered structure of magnitudes, not a description of how a process is performed.

The model therefore provides structural discreteness of magnitudes, while leaving procedures, dynamics, and trajectories outside the scope of the present paper.

15 Scope of the Paper

This paper concerns the algebraic organization of admissible magnitudes of real change above the minimal threshold.

It does not introduce or analyze procedures, trajectories, orientations, operators, variables, dependencies, probabilities, space, time, or dynamical evolution.

It also does not address semantic or reduction-theoretic questions about which variables are real or fictitious in a given structure.

The paper studies only the finitely generated structural model of magnitudes:

$$\mathcal{S} = \{n\delta s : n \in \mathbb{N}_0\}.$$

Further structures, if introduced, belong to later stages of the theory.

16 Conclusion

The Axiom of Minimal Reality establishes the minimal threshold:

$$\delta s = \min \mathcal{S}_{>0}.$$

This threshold alone does not determine the structure of all admissible magnitudes above it.

The present paper studied the finitely generated structural model, in which every admissible magnitude of real change is a finite composition of the minimal real magnitude:

$$\forall x \in \mathcal{S} \quad \exists! n \in \mathbb{N}_0 \quad x = n\delta s.$$

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Under the Finite Generation Principle,

$$\mathcal{S} = \{0, \delta s, 2\delta s, 3\delta s, \dots\}.$$

When the order on $(\mathcal{S})$ is additionally required to be compatible with the generated levels,

$$n\delta s < m\delta s \quad \Longleftrightarrow \quad n < m,$$

one obtains

$$(\mathcal{S}, +, 0, <) \cong (\mathbb{N}_0, +, 0, <).$$

Finite generation is not a second axiom of Minimal Reality. It is a structural model above the minimal threshold.

The threshold is established first. The generated structure is the next step.

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Paper III

Real and Fictitious Variables, Variant 11 and the Principle of Real Dependence

Piotr Dariusz Wójcik

Version 1.3

Abstract

This paper introduces the distinction between real and fictitious variables relative to a given function. A variable is real if changing it, while all other variables are fixed, can change the value of the function; it is fictitious if changing it cannot. The paper formulates the principle of real dependence and presents Variant 11,

$$f(p, q) = \neg q,$$

as the minimal Boolean model separating a real variable from a fictitious one. In this model, q is real and p is fictitious:

$$\mathrm{Real}(f) = \{q\}, \qquad \mathrm{Fict}(f) = \{p\}.$$

A minimal dependence count is introduced by

$$\mathrm{PE}(f) := |\,\mathrm{Real}(f)|,$$

used here only as the number of real variables.

Keywords: real variable, fictitious variable, real dependence, Variant 11, Boolean function, minimal dependence count.

1 Introduction

The first paper in this sequence introduced the Axiom of Minimal Reality as a threshold condition for admissible magnitudes of real change:

$$\delta s = \min \mathcal{S}_{>0}.$$

The second paper studied a structural model of magnitudes above that threshold, in which admissible magnitudes are finite compositions of δs:

$$\forall x \in \mathcal{S} \quad \exists! n \in \mathbb{N}_0 \quad x = n\delta s.$$

The present paper moves to a different but related level of analysis. This transition is conceptual and genealogical rather than deductive: neither the threshold axiom nor the

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finitely generated model is asserted to imply a theory of variable dependence. The present paper does not ask what the minimal magnitude of real change is, nor how magnitudes above that threshold are structurally organized. It asks instead:

When does a variable really participate in the determination of a function?

A variable may occur among the formal arguments of a function without affecting its value. In such a case, its occurrence is syntactic but not real.

The purpose of this paper is to distinguish syntactic occurrence from real dependence.

A variable is real relative to a function if changing it, while all other variables are fixed, can change the value of that function. A variable is fictitious if changing it cannot.

The present paper introduces no new magnitudes, thresholds, or structural assumptions; it concerns only semantic dependence.

This distinction is formalized below, and then illustrated by Variant 11, the Boolean function

$$f(p, q) = \neg q.$$

Variant 11 is treated here as the minimal model of a function with one real variable and one fictitious variable.

2 Syntactic Occurrence and Real Dependence

Let a function be written as

$$f(x_1, \ldots, x_n).$$

The fact that a symbol x_i occurs among the formal arguments of f does not by itself imply that f really depends on x_i.

For example, one may write a function as

$$f(x, y) = x^2.$$

The variable y occurs syntactically as an argument, but changing y never changes the value of f. For all y_1, y_2,

$$f(x, y_1) = f(x, y_2).$$

Thus y is not a real variable of f. It is fictitious relative to f.

This motivates the central distinction of the paper:

$$\boxed{\text{syntactic occurrence} \neq \text{real dependence.}}$$

A variable is real not because it appears in notation, but because its variation, with all other variables fixed, can affect the value of the function.

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3 Formal Setting

Let

$$f : X_1 \times X_2 \times \cdots \times X_n \to Y$$

be a function, and write

$$X := X_1 \times X_2 \times \cdots \times X_n.$$

Elements of X will be denoted by

$$u = (u_1, \ldots, u_n), \qquad v = (v_1, \ldots, v_n).$$

For each index $i \in \{1, \ldots, n\}$, we write

$$u \sim_i v$$

if u and v agree on all coordinates except possibly the i-th coordinate. That is,

$$u \sim_i v \quad \Longleftrightarrow \quad \forall j \neq i,\ u_j = v_j.$$

The relation $u \sim_i v$ therefore means that the only coordinate allowed to vary between u and v is the coordinate i.

This notation allows us to define real and fictitious variables in a compact way.

4 Real Variables

Definition 4.1 (Real variable)**.** Let

$$f : X_1 \times \cdots \times X_n \to Y.$$

The variable x_i is real relative to f if there exist $u, v \in X_1 \times \cdots \times X_n$ such that

$$u \sim_i v$$

and

$$f(u) \neq f(v).$$

The set of all real variables of f is denoted by

$$\mathrm{Real}(f).$$

Remark 4.2 (Classical Terminology)*.* At the formal level used here, a real variable of (f) is exactly what is classically called an *essential variable*: a variable whose variation, with all other variables fixed, can change the value of the function.

Correspondingly, a fictitious variable is an *inessential* or *dummy variable*.

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Thus the terminology

$$\mathrm{Real}(f), \qquad \mathrm{Fict}(f)$$

does not define a different dependence relation from the classical one. The terminology used in this volume records the role assigned to this distinction in the subsequent development of the theory.

Equivalently,

$$x_i \in \mathrm{Real}(f)$$

if and only if

$$\exists u, v \, [u \sim_i v \wedge f(u) \neq f(v)] \, .$$

Expanding the definition of $u \sim_i v$, this becomes

$$x_i \in \mathrm{Real}(f) \quad \Longleftrightarrow \quad \exists u, v \, [(\forall j \neq i, \; u_j = v_j) \wedge f(u) \neq f(v)] \, .$$

Thus x_i is real relative to f precisely when a change of x_i, with all other variables fixed, can change the value of f.

In this sense, realness of a variable is not a property of the symbol alone. It is a property of the role played by that variable in the determination of the function.

5 Fictitious Variables

Definition 5.1 (Fictitious variable)**.** Let

$$f : X_1 \times \cdots \times X_n \to Y.$$

The variable x_i is fictitious relative to f if it is not real relative to f.

The set of fictitious variables of f is denoted by

$$\mathrm{Fict}(f).$$

Thus,

$$\mathrm{Fict}(f) = \{x_1, \ldots, x_n\} \setminus \mathrm{Real}(f).$$

Equivalently,

$$x_i \in \mathrm{Fict}(f)$$

if and only if for all $u, v \in X$,

$$u \sim_i v \quad \Longrightarrow \quad f(u) = f(v).$$

Expanding the definition of $u \sim_i v$, we obtain

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$$x_i \in \mathrm{Fict}(f) \quad \Longleftrightarrow \quad \forall u, v \, [(\forall j \neq i, \; u_j = v_j) \Rightarrow f(u) = f(v)] \, .$$

A fictitious variable may occur in the notation of f, but its change does not affect the value or determination of f.

6 The Principle of Real Dependence

The preceding definitions can be summarized as a general principle.

Principle 6.1 (Real Dependence)**.** *A variable is real relative to a function if and only if changing that variable, while all other variables are fixed, can change the value of the function.*

$$x_i \in \mathrm{Real}(f) \quad \Longleftrightarrow \quad \exists u, v \, [u \sim_i v \wedge f(u) \neq f(v)] \, .$$

Thus real dependence is always relative to a given function. A variable is not declared real merely by appearing in notation. It is real when its variation, with all other variables fixed, can affect the value of the function.

Similarly,

$$x_i \in \mathrm{Fict}(f) \quad \Longleftrightarrow \quad \forall u, v \, [u \sim_i v \Rightarrow f(u) = f(v)] \, .$$

A fictitious variable is therefore a variable whose variation, with all other variables fixed, leaves the value of the function unchanged.

7 Variant 11 as the Minimal Model

We now introduce Variant 11 as the minimal Boolean model separating a real variable from a fictitious variable.

Let

$$p, q \in \{0, 1\}.$$

Define

$$f : \{0, 1\}^2 \to \{0, 1\}$$

by

$$f(p, q) = \neg q.$$

Remark 7.1. The values 0 and 1 are not interpreted as real or fictitious. The real/fictitious distinction applies to variables relative to a function, not to values.

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The variables p and q are classified according to their role in the determination of f.

The function is written with two arguments:

$$f(p, q),$$

but its value depends only on q. This makes Variant 11 the simplest case in which syntactic occurrence and real dependence separate.

Variant 11 is minimal in the sense that it uses two Boolean variables and realizes the smallest nontrivial separation between syntactic occurrence and real dependence: one variable is real and one variable is fictitious.

8 Truth Table of Variant 11

The truth table of

$$f(p, q) = \neg q$$

is

p	q	$f(p, q) = \neg q$
0	0	1
0	1	0
1	0	1
1	1	0

The table is used only to display the dependence structure of the function. The values 0 and 1 are not assigned the status of real or fictitious objects.

The real/fictitious distinction applies to the variables p and q relative to f.

9 Dependence Analysis of Variant 11

We analyze the dependence of f on each variable.

9.1 The variable p

For every $q \in \{0, 1\}$,

$$f(0, q) = \neg q,$$

and

$$f(1, q) = \neg q.$$

Therefore,

$$f(0, q) = f(1, q) \qquad \text{for every } q \in \{0, 1\}.$$

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Changing p, while keeping q fixed, never changes the value of f. Hence p is fictitious relative to f:

$$p \in \mathrm{Fict}(f).$$

9.2 The variable q

For every $p \in \{0,1\}$,

$$f(p,0) = 1,$$

whereas

$$f(p,1) = 0.$$

Therefore,

$$f(p,0) \neq f(p,1) \qquad \text{for every } p \in \{0,1\}.$$

Changing q, while keeping p fixed, changes the value of f. Hence q is real relative to f:

$$q \in \mathrm{Real}(f).$$

Proposition 9.1 (Dependence split of Variant 11)**.** *Let* $f : \{0,1\}^2 \to \{0,1\}$ *be given by*

$$f(p,q) = \neg q.$$

Then

$$\mathrm{Real}(f) = \{q\}, \qquad \mathrm{Fict}(f) = \{p\}.$$

Proof. For every $q \in \{0,1\}$, one has

$$f(0,q) = f(1,q) = \neg q.$$

Thus changing p, with q fixed, never changes the value of f, so $p \in \mathrm{Fict}(f)$.

For every $p \in \{0,1\}$, one has

$$f(p,0) = 1, \qquad f(p,1) = 0.$$

Thus changing q, with p fixed, changes the value of f, so $q \in \mathrm{Real}(f)$. Since the only variables are p and q, the stated equalities follow. □

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10 Meaning of Variant 11

Variant 11 shows, in the simplest Boolean form, that syntactic arity and real dependence need not coincide.

The function

$$f : \{0,1\}^2 \to \{0,1\}$$

is syntactically a function of two variables:

$$p \quad \text{and} \quad q.$$

However, its real dependence is one-dimensional:

$$\mathrm{Real}(f) = \{q\}.$$

The variable p occurs in the expression

$$f(p,q),$$

but does not determine the value of the function. It is therefore fictitious relative to f.

The variable q does determine the value of the function. It is therefore real relative to f.

Thus Variant 11 realizes the central thesis of the paper:

$$\boxed{\text{real dependence is not syntactic occurrence.}}$$

It also provides the minimal model of the distinction:

$$\boxed{\text{one fictitious variable and one real variable.}}$$

11 A Minimal Count of Real Variables

It is sometimes useful to count how many variables of a function are real relative to that function.

In this paper, this count is introduced only in the minimal sense of counting real variables. No further projective, procedural, or dynamic structure is assumed.

Definition 11.1 (Minimal dependence count)**.** For a function f depending on finitely many variables, define

$$\mathrm{PE}(f) := |\,\mathrm{Real}(f)|.$$

In this paper, $\mathrm{PE}(f)$ means only the number of real variables of f. It measures the size of the set

$$\mathrm{Real}(f).$$

For Variant 11,

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$$\mathrm{Real}(f) = \{q\}.$$

Therefore,

$$\mathrm{PE}(f) = 1.$$

This value records that the function has exactly one real variable, although it is syntactically written with two variables.

12 Relation to Papers I and II

The first two papers concern admissible magnitudes of real change.

The first paper establishes the minimal threshold:

$$\delta s = \min \mathcal{S}_{>0}.$$

The second paper studies the finitely generated structural model above that threshold:

$$\forall x \in \mathcal{S} \quad \exists! n \in \mathbb{N}_0 \quad x = n\delta s.$$

The present paper concerns a different question: real dependence within a function.

The distinction can be summarized as follows.

The symbol

$$\delta s$$

answers the question:

what is the minimal real magnitude of change?

The set

$$\mathrm{Real}(f)$$

answers the question:

which variables really participate in the determination of f?

Thus Paper III does not modify the threshold axiom and does not modify the finitely generated model of magnitudes. It introduces a distinct level of analysis: real dependence of variables relative to a function.

The transition from Papers I–II to Paper III is therefore a transition in the development of the theory, not a logical implication from the magnitude structure to the dependence structure.

Variant 11 is the minimal model of this dependence distinction.

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13 Scope of the Paper

The scope of this paper is limited to the distinction between real and fictitious variables relative to a function.

Variant 11 is used only as the minimal Boolean model of this distinction:

$$f(p, q) = \neg q.$$

The paper also introduces the minimal dependence count

$$\mathrm{PE}(f) = |\operatorname{Real}(f)|$$

only as the number of real variables of f.

No additional structure is assumed.

The paper therefore concerns the following objects:

$$\operatorname{Real}(f), \qquad \operatorname{Fict}(f), \qquad f(p, q) = \neg q, \qquad \mathrm{PE}(f) = |\operatorname{Real}(f)|.$$

14 Conclusion

This paper introduced the distinction between real and fictitious variables relative to a function.

A variable is real if changing it, while all other variables are fixed, can change the value of the function:

$$x_i \in \operatorname{Real}(f) \quad \Longleftrightarrow \quad \exists u, v \left[u \sim_i v \wedge f(u) \neq f(v) \right].$$

A variable is fictitious if changing it, while all other variables are fixed, cannot change the value of the function:

$$x_i \in \operatorname{Fict}(f) \quad \Longleftrightarrow \quad \forall u, v \left[u \sim_i v \Rightarrow f(u) = f(v) \right].$$

Variant 11,

$$f(p, q) = \neg q,$$

is the minimal Boolean model of this distinction. In this model,

$$\operatorname{Real}(f) = \{q\},$$

and

$$\operatorname{Fict}(f) = \{p\}.$$

The minimal dependence count is

$$\mathrm{PE}(f) = |\operatorname{Real}(f)|,$$

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and for Variant 11,

$$\mathrm{PE}(f) = 1.$$

The central conclusion is:

> a variable is real when its variation can change the value of the function.

Paper III therefore moves from real magnitude to real dependence.

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Paper IV

Real Support and Canonical Reduction, Reduction of Functions to Their Real Variables

Piotr Dariusz Wójcik

Version 1.3

Abstract

This paper develops the notion of real variables introduced in the previous work. For a function

$$f : X_1 \times \cdots \times X_n \to Y,$$

the set of real variables is denoted by

$$\mathrm{Real}(f).$$

The corresponding real coordinate support is denoted by

$$R_f := \{\, i \in \{1, \ldots, n\} : x_i \in \mathrm{Real}(f) \,\}.$$

The paper proves that f canonically factors through the projection onto this real coordinate support:

$$f = f_{R_f} \circ \pi_{R_f}.$$

Thus fictitious variables may be removed without loss of determination. The paper also proves that R_f is the least coordinate support of f: if f factors through a set of coordinates S, then

$$R_f \subseteq S.$$

The reduction considered here is purely static and structural; it is a factorization through a projection, not a procedure, trajectory, collapse, or dynamic process.

Keywords: real support, real variables, fictitious variables, canonical reduction, factorization, projection.

1 Introduction

The previous paper introduced the distinction between real and fictitious variables relative to a function. For a function

$$f : X_1 \times \cdots \times X_n \to Y,$$

a variable is real if changing it can change the value of f, while all other variables are kept fixed. A variable is fictitious if changing it cannot change the value of f.

The present paper develops the next static consequence of that distinction.

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If the real variables of f are known, then the fictitious variables should be removable without loss of determination. In other words, the function should be reducible to its real variables.

The purpose of this paper is to show that a function canonically factors through its real variables.

Let

$$R_f := \{\, i \in \{1, \dots, n\} : x_i \in \mathrm{Real}(f) \,\}.$$

Let

$$X_{R_f} := \prod_{i \in R_f} X_i$$

be the coordinate space corresponding to the real variables, and let

$$\pi_{R_f} : X_1 \times \cdots \times X_n \to X_{R_f}$$

be the projection onto those coordinates.

The main result states that there exists a unique function

$$f_{R_f} : X_{R_f} \to Y$$

such that

$$f = f_{R_f} \circ \pi_{R_f}.$$

This is the canonical reduction of f to its real variables.

The reduction considered in this paper is static: it is a factorization through a projection, not a process.

2 Reminder: Real and Fictitious Variables

Let

$$f : X_1 \times \cdots \times X_n \to Y$$

be a function. Throughout the paper we assume that

$$X_i \neq \varnothing \qquad \text{for every } i = 1, \dots, n.$$

Let

$$X := X_1 \times \cdots \times X_n.$$

For $u, v \in X$ and $i \in \{1, \dots, n\}$, write

$$u \sim_i v$$

if u and v agree on all coordinates except possibly the i-th coordinate. Formally,

$$u \sim_i v \quad \iff \quad \forall j \neq i,\ u_j = v_j.$$

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The variable x_i is real relative to f if

$$\exists u, v \in X \left[u \sim_i v \wedge f(u) \neq f(v) \right].$$

Thus,

$$x_i \in \mathrm{Real}(f) \quad \Longleftrightarrow \quad \exists u, v \in X \left[u \sim_i v \wedge f(u) \neq f(v) \right].$$

The variable x_i is fictitious relative to f if changing it never changes the value of f. Equivalently,

$$x_i \in \mathrm{Fict}(f) \quad \Longleftrightarrow \quad \forall u, v \in X \left[u \sim_i v \Rightarrow f(u) = f(v) \right].$$

The two sets of variables are complementary:

$$\mathrm{Fict}(f) = \{x_1, \ldots, x_n\} \setminus \mathrm{Real}(f).$$

3 Real Support

Definition 3.1 (Real support)**.** Let

$$f : X_1 \times \cdots \times X_n \to Y.$$

The real support of f is the set of coordinates corresponding to the real variables of f:

$$R_f := \{\, i \in \{1, \ldots, n\} : x_i \in \mathrm{Real}(f) \,\}.$$

The set R_f is not the set of coordinates syntactically appearing in the expression for f. It is the set of coordinates whose associated variables actually participate in the determination of f.

Thus

$$R_f$$

records the coordinates that cannot be changed freely without possibly changing the value of f.

Equivalently,

$$\mathrm{Real}(f) = \{\, x_i : i \in R_f \,\}.$$

The fictitious variables are exactly the complementary variables:

$$\mathrm{Fict}(f) = \{x_1, \ldots, x_n\} \setminus \mathrm{Real}(f).$$

The coordinates outside R_f are precisely those whose associated variables are fictitious.

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4 Coordinate Spaces and Projections

For every subset

$$S \subseteq \{1, \dots, n\},$$

define

$$X_S := \prod_{i \in S} X_i.$$

Products over subsets of $\{1, \dots, n\}$ are taken with the inherited order of indices.
If

$$S = \varnothing,$$

we use the standard convention

$$X_\varnothing := \{*\}.$$

For each subset $S \subseteq \{1, \dots, n\}$, let

$$\pi_S : X \to X_S$$

denote the projection onto the coordinates indexed by S:

$$\pi_S(x_1, \dots, x_n) = (x_i)_{i \in S}.$$

In particular, for the real support R_f, we have the coordinate space

$$X_{R_f} := \prod_{i \in R_f} X_i$$

and the projection

$$\pi_{R_f} : X \to X_{R_f}.$$

The projection π_{R_f} keeps the real coordinates and forgets the fictitious coordinates.
It is important that this is only a static projection. It is not a procedure and not a process.

Remark 4.1. If $R_f = \varnothing$, then f has no real variables. In that case f is constant on X, and its reduction factors through the unique map

$$\pi_\varnothing : X \to \{*\}.$$

5 The Reduced Function

The goal is to define a function

$$f_{R_f} : X_{R_f} \to Y$$

such that

$$f = f_{R_f} \circ \pi_{R_f}.$$

The natural definition is the following.
For

$$r \in X_{R_f},$$

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choose any point

$$x \in X$$

such that

$$\pi_{R_f}(x) = r.$$

Then define

$$f_{R_f}(r) := f(x).$$

The only issue is well-definedness. We must show that if two points $x, y \in X$ have the same real coordinates, then they have the same value under f.

This is the content of the next lemma.

6 Fictitious Coordinates Can Be Changed Simultaneously

Lemma 6.1 (Simultaneous change of fictitious coordinates)**.** *Let R_f be the real coordinate support of f. If $x, y \in X$ satisfy*

$$\pi_{R_f}(x) = \pi_{R_f}(y),$$

then

$$f(x) = f(y).$$

Proof. The equality

$$\pi_{R_f}(x) = \pi_{R_f}(y)$$

means that x and y agree on all coordinates belonging to R_f.

Thus x and y can differ only on coordinates outside

$$R_f.$$

Let

$$i_1, \ldots, i_k$$

be the finite list of coordinates on which x and y differ. Each of these coordinates lies outside R_f, so each associated variable x_{i_m} is fictitious relative to f.

Construct a finite sequence

$$x = x^{(0)}, x^{(1)}, \ldots, x^{(k)} = y$$

such that $x^{(m)}$ is obtained from $x^{(m-1)}$ by changing only the coordinate i_m from its value in x to its value in y.

For each $m = 1, \ldots, k$, the points $x^{(m-1)}$ and $x^{(m)}$ differ only on the coordinate i_m. Since the associated variable x_{i_m} is fictitious relative to f, we have

$$f(x^{(m-1)}) = f(x^{(m)}).$$

Therefore,

$$f(x) = f(x^{(0)}) = f(x^{(1)}) = \cdots = f(x^{(k)}) = f(y).$$

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Hence

$$f(x) = f(y).$$

□

7 Canonical Reduction Theorem

Theorem 7.1 (Canonical Reduction)**.** *Let*

$$f : X_1 \times \cdots \times X_n \to Y$$

and let R_f be the real coordinate support of f. Then there exists a unique function

$$f_{R_f} : X_{R_f} \to Y$$

such that

$$\boxed{f = f_{R_f} \circ \pi_{R_f}.}$$

Proof. Let

$$r \in X_{R_f}.$$

Since all X_i are nonempty, there exists $x \in X$ such that

$$\pi_{R_f}(x) = r.$$

Define

$$f_{R_f}(r) := f(x).$$

We first show that this definition is independent of the chosen representative x.
Suppose that $x, y \in X$ both satisfy

$$\pi_{R_f}(x) = r \qquad \text{and} \qquad \pi_{R_f}(y) = r.$$

Then

$$\pi_{R_f}(x) = \pi_{R_f}(y).$$

By the previous lemma,

$$f(x) = f(y).$$

Hence f_{R_f} is well-defined.
Now let $x \in X$. By definition of f_{R_f},

$$f_{R_f}(\pi_{R_f}(x)) = f(x).$$

Therefore

$$f = f_{R_f} \circ \pi_{R_f}.$$

It remains to prove uniqueness.

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Suppose that

$$g : X_{R_f} \to Y$$

also satisfies

$$f = g \circ \pi_{R_f}.$$

Let

$$r \in X_{R_f}.$$

Choose $x \in X$ such that

$$\pi_{R_f}(x) = r.$$

Then

$$g(r) = g(\pi_{R_f}(x)) = f(x).$$

On the other hand, by the definition of f_{R_f},

$$f_{R_f}(r) = f(x).$$

Thus

$$g(r) = f_{R_f}(r).$$

Since $r \in X_{R_f}$ was arbitrary,

$$g = f_{R_f}.$$

Therefore the reduced function is unique. □

Remark 7.2. The reduction is canonical because the reduced function f_{R_f} is uniquely determined by f and its real support R_f. No additional choice of structure is involved.

Corollary 7.3. *The value of f is completely determined by its real-coordinate projection.*

Proof. By the Canonical Reduction Theorem,

$$f = f_{R_f} \circ \pi_{R_f}.$$

Thus the value of $f(x)$ is determined by $\pi_{R_f}(x)$, that is, by the real coordinates of x. □

8 The Least Coordinate Support

The previous theorem shows that f factors through its real support. We now show that the real support is the least coordinate support through which f can factor.

Let

$$S \subseteq \{1, \ldots, n\}.$$

Recall that

$$X_S = \prod_{i \in S} X_i$$

and that

$$\pi_S : X \to X_S$$

is the projection onto the coordinates in S.

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Theorem 8.1 (Least coordinate support)**.** *If there exists a function*

$$g : X_S \to Y$$

such that

$$f = g \circ \pi_S,$$

then

$$\boxed{R_f \subseteq S.}$$

Proof. Suppose, for contradiction, that

$$R_f \not\subseteq S.$$

Then there exists

$$i \in R_f$$

such that

$$i \notin S.$$

Since $i \in R_f$, the variable x_i belongs to $\mathrm{Real}(f)$. Therefore there exist $u, v \in X$ such that

$$u \sim_i v$$

and

$$f(u) \neq f(v).$$

The relation $u \sim_i v$ means that u and v differ only possibly on the coordinate i. Since $i \notin S$, the projections of u and v onto S are equal:

$$\pi_S(u) = \pi_S(v).$$

Using the assumed factorization

$$f = g \circ \pi_S,$$

we obtain

$$f(u) = g(\pi_S(u))$$

and

$$f(v) = g(\pi_S(v)).$$

Since

$$\pi_S(u) = \pi_S(v),$$

it follows that

$$f(u) = f(v).$$

This contradicts the choice of u and v. Therefore

$$R_f \subseteq S.$$

□

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Corollary 8.2. *The real support R_f is the least coordinate support through which f factors.*

Proof. By the Canonical Reduction Theorem, f factors through R_f:

$$f = f_{R_f} \circ \pi_{R_f}.$$

By the Least Coordinate Support Theorem, if f factors through any coordinate set S, then

$$R_f \subseteq S.$$

Hence R_f is the least coordinate support through which f factors. □

Thus R_f is not merely a minimal coordinate support, but the least coordinate support through which f factors.

No real coordinate can be omitted from any coordinate factorization of f.

Fictitious variables may be removed without loss of determination, while real variables cannot be removed without possible loss of determination.

9 Variant 11 as a Canonical Minimal Example

We now illustrate canonical reduction using Variant 11.

Let

$$p, q \in \{0, 1\}$$

and define

$$f : \{0, 1\}^2 \to \{0, 1\}$$

by

$$f(p, q) = \neg q.$$

From the previous analysis of Variant 11, we have

$$\text{Real}(f) = \{q\}$$

and

$$\text{Fict}(f) = \{p\}.$$

Identifying p with the first coordinate and q with the second coordinate, the real coordinate support is

$$R_f = \{2\}.$$

The coordinate space of the real support is

$$X_{R_f} = \{0, 1\}.$$

The projection onto the real support is

$$\pi_{R_f}(p, q) = q.$$

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The reduced function is

$$f_{R_f} : \{0,1\} \to \{0,1\}$$

defined by

$$f_{R_f}(q) = \neg q.$$

For every $(p, q) \in \{0,1\}^2$,

$$f_{R_f}(\pi_{R_f}(p,q)) = f_{R_f}(q) = \neg q = f(p,q).$$

Therefore

$$f = f_{R_f} \circ \pi_{R_f}.$$

This shows that the fictitious variable p may be removed without loss of determination, while the real variable q remains necessary.

Indeed, if f factors through a coordinate set S, then by the least coordinate support theorem,

$$R_f \subseteq S.$$

Since

$$R_f = \{2\},$$

we must have

$$2 \in S.$$

Equivalently, the coordinate corresponding to q must belong to S.

Thus q cannot be removed, while p can be removed.

Variant 11 is a canonical minimal example of reduction to one real variable.

10 Relation to Paper III

Paper III introduced the distinction between real and fictitious variables:

$$\text{Real}(f), \qquad \text{Fict}(f).$$

Paper IV uses that distinction to obtain the canonical reduction:

$$f = f_{R_f} \circ \pi_{R_f}.$$

The relation is therefore:

Paper III identifies the real variables.

Paper IV reduces the function to those variables.

In symbolic form:

$$\text{Real}(f) \quad \longrightarrow \quad R_f \quad \longrightarrow \quad f = f_{R_f} \circ \pi_{R_f}.$$

Paper IV does not modify the definitions of real and fictitious variables. It uses them to prove that fictitious variables can be removed without loss of determination.

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11 Scope of the Paper

The scope of this paper is limited to static canonical reduction.

The objects considered are:

$$R_f := \{\, i \in \{1, \ldots, n\} : x_i \in \mathrm{Real}(f) \,\},$$

$$X_{R_f} = \prod_{i \in R_f} X_i,$$

$$\pi_{R_f} : X \to X_{R_f},$$

$$f_{R_f} : X_{R_f} \to Y,$$

and

$$f = f_{R_f} \circ \pi_{R_f}.$$

The paper also establishes the least-coordinate-support condition:

$$f = g \circ \pi_S \quad \Longrightarrow \quad R_f \subseteq S.$$

The reduction considered here is the static factorization

$$f = f_{R_f} \circ \pi_{R_f}.$$

12 Conclusion

This paper established the canonical reduction of a function to its real variables.

If

$$R_f := \{\, i \in \{1, \ldots, n\} : x_i \in \mathrm{Real}(f) \,\},$$

then there exists a unique function

$$f_{R_f} : X_{R_f} \to Y$$

such that

$$\boxed{f = f_{R_f} \circ \pi_{R_f}.}$$

Thus fictitious variables may be removed without loss of determination.

The paper also proved that R_f is the least coordinate support of f: if

$$f = g \circ \pi_S,$$

then

$$\boxed{R_f \subseteq S.}$$

Hence the real coordinate support consists exactly of the coordinates that cannot be removed from any coordinate factorization of f.

For Variant 11,

$$f(p, q) = \neg q,$$

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with p as the first coordinate and q as the second coordinate, we have

$$R_f = \{2\},$$

$$\pi_{R_f}(p, q) = q,$$

$$f_{R_f}(q) = \neg q,$$

and therefore

$$scope f = f_{R_f} \circ \pi_{R_f}.$$

The central conclusion is:

fictitious variables can be removed without loss of determination.

The reduction is static and canonical.

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Paper V

Procedural Reduction to Real Support. Unit Admissible Steps from a Function to Its Canonical Reduction

Piotr Dariusz Wójcik

Version 1.3

Abstract

This paper develops the first procedural level of canonical reduction. The previous paper established the static factorization

$$f = f_{R_f} \circ \pi_{R_f},$$

where R_f is the real coordinate support of f. The present paper studies how this reduction can be realized by a finite sequence of unit admissible steps. Each step removes exactly one coordinate outside the real support, equivalently one currently fictitious coordinate.

To avoid ambiguities of reindexing, reduction states are represented by pairs

$$(K, f_K),$$

where $K \subseteq \{1, \dots, n\}$ is the set of remaining original coordinates and

$$f = f_K \circ \pi_K.$$

The paper proves that every function admits a complete unit admissible reduction procedure from

$$(\{1, \dots, n\}, f)$$

to

$$(R_f, f_{R_f}),$$

and that every complete unit reduction procedure has length

$$|F_f|,$$

where

$$F_f := \{1, \dots, n\} \setminus R_f$$

is the coordinate support corresponding to the fictitious variables of f. Thus, within the class of complete unit reduction procedures, the minimal length is

$$\ell_{\min}(f) = |F_f|.$$

Keywords: procedural reduction, real support, fictitious variables, admissible step, unit procedure, canonical reduction.

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1 Introduction

The previous paper established the static canonical reduction of a function to its real variables. For a function

$$f : X_1 \times \cdots \times X_n \to Y,$$

with real coordinate support

$$R_f := \{\, i \in \{1, \ldots, n\} : x_i \in \mathrm{Real}(f) \,\},$$

there exists a unique reduced function

$$f_{R_f} : X_{R_f} \to Y$$

such that

$$f = f_{R_f} \circ \pi_{R_f}.$$

This result says that the fictitious variables of f may be removed without loss of determination. It is a static result: it describes the factorization of f through its real support.

The present paper asks a new question:

How can this canonical reduction be realized step by step?

The purpose of this paper is to describe the procedural realization of canonical reduction by unit admissible steps.

A unit step removes exactly one coordinate outside the real support, equivalently one fictitious coordinate relative to the current reduced function. A complete unit reduction procedure starts from the original function and ends at the canonical reduction:

$$(\{1, \ldots, n\}, f) \longrightarrow \cdots \longrightarrow (R_f, f_{R_f}).$$

The paper is limited to finite unit reduction procedures, their completeness, and their length.

2 Reminder: Canonical Reduction

Let

$$f : X_1 \times \cdots \times X_n \to Y$$

be a function. Throughout the paper we assume that

$$X_i \neq \varnothing \qquad \text{for every } i = 1, \ldots, n.$$

Let

$$N := \{1, \ldots, n\}$$

and

$$X := X_1 \times \cdots \times X_n.$$

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The real coordinate support of f is

$$R_f := \{\, i \in N : x_i \in \mathrm{Real}(f) \,\}.$$

The corresponding fictitious coordinate support is

$$F_f := N \setminus R_f.$$

Thus F_f consists precisely of the original coordinates whose variables are fictitious relative to f.

For every subset

$$K \subseteq N,$$

define

$$X_K := \prod_{i \in K} X_i.$$

Products over subsets of N are taken with the inherited order of indices.

If

$$K = \varnothing,$$

we use the standard convention

$$X_\varnothing := \{*\}.$$

For

$$L \subseteq K \subseteq N,$$

let

$$\pi_L^K : X_K \to X_L$$

denote the projection from X_K onto the coordinates indexed by L.

In particular, for $K \subseteq N$, we write

$$\pi_K : X \to X_K$$

for the projection from the full coordinate space to X_K.

These projections satisfy the compatibility relation

$$\pi_L = \pi_L^K \circ \pi_K \qquad \text{whenever } L \subseteq K \subseteq N.$$

The canonical reduction theorem from the previous paper states that there exists a unique function

$$f_{R_f} : X_{R_f} \to Y$$

such that

$$\boxed{f = f_{R_f} \circ \pi_{R_f}.}$$

The present paper treats f_{R_f} as the target of a finite unit reduction procedure.

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3 Intermediate Reductions

Before defining procedures, we establish the intermediate reductions through coordinate sets that still contain the real support.

Lemma 3.1 (Intermediate reduction)**.** *Let*

$$K \subseteq N$$

satisfy

$$R_f \subseteq K \subseteq N.$$

Then there exists a unique function

$$f_K : X_K \to Y$$

such that

$$\boxed{f = f_K \circ \pi_K.}$$

Proof. Since

$$R_f \subseteq K,$$

we have the projection

$$\pi^K_{R_f} : X_K \to X_{R_f}.$$

Define

$$f_K := f_{R_f} \circ \pi^K_{R_f}.$$

Then, using

$$\pi_{R_f} = \pi^K_{R_f} \circ \pi_K,$$

we obtain

$$f_K \circ \pi_K = f_{R_f} \circ \pi^K_{R_f} \circ \pi_K = f_{R_f} \circ \pi_{R_f} = f.$$

Thus such a function f_K exists.
To prove uniqueness, suppose that

$$g : X_K \to Y$$

also satisfies

$$f = g \circ \pi_K.$$

Let

$$u \in X_K.$$

Since all X_i are nonempty, there exists $x \in X$ such that

$$\pi_K(x) = u.$$

Then

$$g(u) = g(\pi_K(x)) = f(x).$$

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On the other hand,

$$f_K(u) = f_K(\pi_K(x)) = f(x).$$

Hence

$$g(u) = f_K(u).$$

Since $u \in X_K$ was arbitrary,

$$g = f_K.$$

Therefore f_K is unique. □

The family of functions

$$f_K : X_K \to Y, \qquad R_f \subseteq K \subseteq N,$$

forms the static background through which unit reduction procedures will move.

In particular,

$$f_N = f$$

and

$$f_{R_f}$$

is the canonical reduction.

4 Reduction States

Definition 4.1 (Reduction state)**.** A reduction state of f is a pair

$$(K, f_K),$$

where

$$R_f \subseteq K \subseteq N$$

and

$$f_K : X_K \to Y$$

is the unique intermediate reduction satisfying

$$f = f_K \circ \pi_K.$$

The initial reduction state is

$$(N, f_N) = (N, f).$$

The terminal canonical reduction state is

$$(R_f, f_{R_f}).$$

A reduction procedure will be a finite sequence of reduction states starting at

$$(N, f)$$

and, if complete, ending at

$$(R_f, f_{R_f}).$$

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5 Real and Fictitious Coordinates in a Reduction State

Let

$$(K, f_K)$$

be a reduction state.

For $u, v \in X_K$ and $i \in K$, write

$$u \sim_i^K v$$

if u and v agree on all coordinates of K except possibly the coordinate i. That is,

$$u \sim_i^K v \quad \Longleftrightarrow \quad \forall j \in K,\ j \neq i, \quad u_j = v_j.$$

The coordinate $i \in K$ is real relative to f_K if

$$\exists u, v \in X_K \left[u \sim_i^K v \wedge f_K(u) \neq f_K(v) \right].$$

The coordinate $i \in K$ is fictitious relative to f_K if changing it never changes the value of f_K. Equivalently, i is fictitious relative to f_K if

$$\forall u, v \in X_K \left[u \sim_i^K v \Rightarrow f_K(u) = f_K(v) \right].$$

6 Support Preservation

We now prove that intermediate reductions preserve exactly the original real coordinate support.

Lemma 6.1 (Support preservation)**.** *Let*

$$(K, f_K)$$

be a reduction state. Then the real coordinates of f_K are precisely R_f, and the fictitious coordinates of f_K are precisely $K \setminus R_f$.

Proof. We prove both statements.

First, let

$$i \in R_f.$$

By the definition of R_f, the variable x_i is real relative to f. Hence there exist $x, y \in X$ such that

$$x \sim_i y$$

and

$$f(x) \neq f(y).$$

Let

$$u = \pi_K(x), \qquad v = \pi_K(y).$$

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Since $R_f \subseteq K$, we have $i \in K$. Because x and y differ only possibly on coordinate i, the projected points u and v differ only possibly on coordinate i inside K. Hence

$$u \sim_i^K v.$$

Moreover,

$$f_K(u) = f_K(\pi_K(x)) = f(x),$$

and

$$f_K(v) = f_K(\pi_K(y)) = f(y).$$

Since

$$f(x) \neq f(y),$$

we obtain

$$f_K(u) \neq f_K(v).$$

Thus i is real relative to f_K. Therefore every coordinate in R_f remains real relative to f_K.

Conversely, let

$$i \in K \setminus R_f.$$

Then the variable x_i is fictitious relative to f. We show that i is fictitious relative to f_K.

Let $u, v \in X_K$ satisfy

$$u \sim_i^K v.$$

Choose lifts $x, y \in X$ such that

$$\pi_K(x) = u, \qquad \pi_K(y) = v,$$

and such that x and y agree on all coordinates outside K. Such lifts exist because all coordinate sets X_j are nonempty.

Since u and v differ only possibly on i, and x, y agree outside K, the points x and y differ only possibly on coordinate i. Hence

$$x \sim_i y.$$

Because x_i is fictitious relative to f, we have

$$f(x) = f(y).$$

Using the factorization $f = f_K \circ \pi_K$, we obtain

$$f_K(u) = f_K(\pi_K(x)) = f(x) = f(y) = f_K(\pi_K(y)) = f_K(v).$$

Thus i is fictitious relative to f_K.

Since every coordinate in K is either in R_f or in $K \setminus R_f$, the real coordinates of f_K are precisely R_f, and the fictitious coordinates of f_K are precisely $K \setminus R_f$. □

Remark 6.2. The lemma says that during a correct reduction, the real coordinates remain exactly the original real coordinate support R_f, while the fictitious coordinates at the state K are precisely the remaining coordinates outside R_f.

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7 Single-Variable Reduction Step

Proposition 7.1 (Single-variable reduction step)**.** *Let*

$$(K, f_K)$$

be a reduction state, and let

$$i \in K \setminus R_f.$$

Set

$$K' := K \setminus \{i\}.$$

Then

$$\boxed{f_K = f_{K'} \circ \pi^K_{K'}.}$$

Consequently, removing the fictitious coordinate i yields a transition between reduction states:

$$(K, f_K) \longrightarrow (K', f_{K'}).$$

Proof. By the support preservation lemma, the coordinate i is fictitious relative to f_K. Since

$$R_f \subseteq K',$$

there exists the intermediate reduction

$$f_{K'} : X_{K'} \to Y$$

satisfying

$$f = f_{K'} \circ \pi_{K'}.$$

We also have

$$f = f_K \circ \pi_K.$$

Since

$$\pi_{K'} = \pi^K_{K'} \circ \pi_K,$$

we obtain

$$f = f_{K'} \circ \pi_{K'} = f_{K'} \circ \pi^K_{K'} \circ \pi_K.$$

By the uniqueness of f_K in the factorization through π_K, it follows that

$$f_K = f_{K'} \circ \pi^K_{K'}.$$

Thus $(K', f_{K'})$ is the reduction state obtained by removing the single fictitious coordinate i. □

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8 Admissible Unit Step

Definition 8.1 (Admissible unit step)**.** Let

$$(K, f_K)$$

be a reduction state. An admissible unit step from (K, f_K) is a transition

$$(K, f_K) \longrightarrow_i (K', f_{K'})$$

such that

$$K' = K \setminus \{i\}$$

and

$$i \in K \setminus R_f.$$

By the support preservation lemma, this is equivalent to saying that i is fictitious relative to the current reduced function f_K.

Thus an admissible unit step removes exactly one currently fictitious coordinate.

It never removes a real coordinate.

9 Unit Reduction Procedures

Definition 9.1 (Unit reduction procedure)**.** A unit reduction procedure for f is a finite sequence of reduction states

$$(K_0, f_{K_0}) \longrightarrow_{i_1} (K_1, f_{K_1}) \longrightarrow_{i_2} \cdots \longrightarrow_{i_k} (K_k, f_{K_k})$$

such that

$$K_0 = N,$$

$$f_{K_0} = f,$$

and for every $m = 1, \ldots, k$,

$$K_m = K_{m-1} \setminus \{i_m\},$$

where

$$i_m \in K_{m-1} \setminus R_f.$$

Each step removes exactly one coordinate outside the real support. Equivalently, by support preservation, each step removes one fictitious coordinate relative to the current state.

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10 Complete Unit Reduction Procedures

Definition 10.1 (Complete unit reduction procedure)**.** A unit reduction procedure

$$(K_0, f_{K_0}) \longrightarrow \cdots \longrightarrow (K_k, f_{K_k})$$

is complete if

$$K_k = R_f.$$

If the procedure is complete, then

$$f_{K_k} = f_{R_f}.$$

Thus a complete unit reduction procedure has the form

$$(N, f) \longrightarrow \cdots \longrightarrow (R_f, f_{R_f}).$$

This formulation avoids any ambiguity of reindexing: every state is indexed by a subset of the original coordinate set N.

11 Existence of Complete Unit Reduction Procedures

Proposition 11.1 (Existence of complete unit reduction procedures)**.** *Every function*

$$f : X_1 \times \cdots \times X_n \to Y$$

admits a complete unit reduction procedure.

Proof. Let

$$F_f = N \setminus R_f.$$

If

$$F_f = \varnothing,$$

then

$$R_f = N.$$

The trivial procedure of length 0,

$$(N, f),$$

is already complete.

Assume now that

$$F_f \neq \varnothing.$$

Choose any ordering

$$F_f = \{i_1, \ldots, i_k\}.$$

Define

$$K_m := N \setminus \{i_1, \ldots, i_m\}$$

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for

$$m = 0, \dots, k.$$

Thus

$$K_0 = N$$

and

$$K_k = N \setminus F_f = R_f.$$

For each m, let

$$f_{K_m} : X_{K_m} \to Y$$

be the intermediate reduction satisfying

$$f = f_{K_m} \circ \pi_{K_m}.$$

We must show that every step

$$(K_{m-1}, f_{K_{m-1}}) \longrightarrow_{i_m} (K_m, f_{K_m})$$

is admissible.

By construction,

$$i_m \in K_{m-1}$$

and

$$K_m = K_{m-1} \setminus \{i_m\}.$$

Moreover, since $i_m \in F_f$, we have

$$i_m \notin R_f.$$

Hence

$$i_m \in K_{m-1} \setminus R_f.$$

Thus

$$i_m \in K_{m-1} \setminus R_f.$$

Therefore each step is admissible.

The resulting sequence

$$(N, f) = (K_0, f_{K_0}) \longrightarrow_{i_1} (K_1, f_{K_1}) \longrightarrow_{i_2} \cdots \longrightarrow_{i_k} (K_k, f_{K_k}) = (R_f, f_{R_f})$$

is a complete unit reduction procedure. □

12 Order Independence of the Terminal Reduction

Remark 12.1 (Order independence of the terminal reduction). Different orderings of the fictitious coordinates may produce different intermediate sequences of states. Every complete unit reduction procedure, however, terminates at the same canonical state

$$(R_f, f_{R_f}).$$

Thus the terminal reduction is independent of the order in which fictitious coordinates are removed; this records the canonicality of the terminal object.

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13 Length of a Unit Procedure

Definition 13.1 (Length)**.** Let

$$P : (K_0, f_{K_0}) \longrightarrow (K_1, f_{K_1}) \longrightarrow \cdots \longrightarrow (K_k, f_{K_k})$$

be a unit reduction procedure. Its length is the number of its steps:

$$\ell(P) := k.$$

Since each unit step removes exactly one coordinate, one has

$$|K_m| = |K_0| - m.$$

14 Length and Minimal Unit Length

In this paper, the symbol

$$\ell_{\min}(f)$$

denotes only the minimal length within the class of complete unit reduction procedures.

Definition 14.1 (Minimal unit length)**.** The minimal unit length of f is defined by

$$\ell_{\min}(f) := \min \{\ell(P) \mid P \text{ is a complete unit reduction procedure for } f\}.$$

Proposition 14.2 (Length of complete unit reduction procedures)**.** *Every complete unit reduction procedure P for f has length*

$$\ell(P) = |F_f|.$$

Consequently,

$$\boxed{\ell_{\min}(f) = |F_f|.}$$

Proof. Let

$$P : (K_0, f_{K_0}) \longrightarrow \cdots \longrightarrow (K_k, f_{K_k})$$

be a complete unit reduction procedure.

Since the procedure is complete,

$$K_0 = N$$

and

$$K_k = R_f.$$

Each unit step removes exactly one coordinate. Therefore

$$\ell(P) = k = |N| - |R_f|.$$

But

$$F_f = N \setminus R_f,$$

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so

$$|F_f| = |N| - |R_f|.$$

Hence

$$\ell(P) = |F_f|.$$

By the proposition on the existence of complete unit reduction procedures, the class over which $\ell_{\min}(f)$ is defined is nonempty. Therefore

$$\ell_{\min}(f) = |F_f|.$$

□

Remark 14.3. The quantity

$$\ell_{\min}(f)$$

counts the number of unit fictitious-coordinate removals needed to reach the canonical reduction. It belongs only to the unit-procedural level studied in this paper.

15 Main Result

Combining the previous results gives the main theorem of the paper.

Theorem 15.1 (Procedural realization of canonical reduction)**.** *Every function*

$$f : X_1 \times \cdots \times X_n \to Y$$

admits a complete unit reduction procedure

$$(N, f) \longrightarrow \cdots \longrightarrow (R_f, f_{R_f}).$$

Every complete unit reduction procedure has length

$$|F_f|.$$

Consequently,

$$\boxed{\ell_{\min}(f) = |F_f|.}$$

Proof. The existence statement follows from the proposition on complete unit reduction procedures. The length statement follows from the preceding proposition. □

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16 Variant 11 as a One-Step Example

We now illustrate the procedure on Variant 11.

Let

$$p, q \in \{0, 1\}$$

and define

$$f : \{0, 1\}^2 o \{0, 1\}$$

by

$$f(p, q) = egq.$$

Here the original coordinate set is

$$N = \{1, 2\},$$

where the first coordinate is labelled by p and the second coordinate is labelled by q.

For this function,

$$\mathrm{Real}(f) = \{q\}$$

and

$$\mathrm{Fict}(f) = \{p\}.$$

Thus the real coordinate support and fictitious coordinate support are

$$R_f = \{2\}, \qquad F_f = \{1\}.$$

The initial reduction state is

$$(\{1, 2\}, f),$$

where

$$f(p, q) = egq.$$

Since

$$1 \in F_f,$$

there is an admissible unit step removing the p-coordinate:

$$(\{1, 2\}, f) \longrightarrow_1 (\{2\}, f_{\{2\}}).$$

The terminal function is

$$f_{\{2\}}(q) = egq.$$

Since

$$\{2\} = R_f,$$

we have

$$f_{\{2\}} = f_{R_f}.$$

Thus the complete procedure is

$$(\{1, 2\}, f) \longrightarrow_1 (\{2\}, f_{R_f}).$$

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Its length is

$$\ell = 1.$$

Since

$$|F_f| = 1,$$

we obtain

$$\ell_{\min}(f) = 1.$$

Variant 11 is therefore a one-step example of a complete unit reduction procedure.

17 Relation to Paper IV

Paper IV established the static canonical reduction

$$f = f_{R_f} \circ \pi_{R_f},$$

where R_f is the real coordinate support associated with the real variables of f.

Paper V gives a procedural realization of this reduction:

$$(N, f) \longrightarrow \cdots \longrightarrow (R_f, f_{R_f}).$$

Thus the relation between the two papers is:

Paper IV establishes the canonical target.

Paper V gives a finite unit procedure reaching that target.

Paper IV is static. Paper V is procedural, but only at the minimal unit level.

18 Scope of the Paper

This paper concerns complete unit admissible procedures of reduction to real support.

It introduces and studies:

reduction states,

unit admissible steps,

unit reduction procedures,

complete unit reduction procedures,

order independence of the terminal reduction,

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length,

and

$$\ell_{\min}(f) = |F_f|.$$

The present paper stops at the level of complete unit reduction procedures and their finite lengths.

19 Conclusion

The previous paper proved that every function canonically factors through its real coordinate support:

$$f = f_{R_f} \circ \pi_{R_f}.$$

The present paper showed that this static canonical reduction can be realized procedurally by a finite sequence of unit admissible steps.

A reduction state has the form

$$(K, f_K),$$

where

$$f = f_K \circ \pi_K.$$

An admissible unit step removes exactly one fictitious coordinate relative to the current state:

$$(K, f_K) \longrightarrow_i (K \setminus \{i\}, f_{K\setminus\{i\}}),$$

where

$$i \in K \setminus R_f.$$

Every function admits a complete unit reduction procedure

$$(N, f) \longrightarrow \cdots \longrightarrow (R_f, f_{R_f}).$$

Every such procedure has length

$$|F_f|.$$

Therefore, within the class of complete unit reduction procedures,

$$\boxed{\ell_{\min}(f) = |F_f|.}$$

The central conclusion is:

canonical reduction can be reached by finitely many unit removals of fictitious variables.

This establishes the finite unit-procedural realization of canonical reduction.

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Paper VI

Variant 11 and the Foundations of Real Support. Necessary Consequences and Technical Preliminaries

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Version 1.3

Abstract

This paper has two distinct parts.
Part I studies Variant 11, the Boolean function

$$f_{11} : \{0,1\}^2 \to \{0,1\}, \qquad f_{11}(p,q) = \neg q.$$

It isolates the direct structural consequences of this function: the separation between syntactic occurrence and real determination, the real/fictitious split

$$\mathrm{Real}(f_{11}) = \{q\}, \qquad \mathrm{Fict}(f_{11}) = \{p\},$$

and the local factorization

$$f_{11} = g \circ \pi_q, \qquad \pi_q(p,q) = q, \qquad g(q) = \neg q.$$

It then places this local structure within the canonical and unit-procedural reduction framework established in Papers III–V. With

$$x_1 = p, \qquad x_2 = q,$$

the corresponding coordinate supports are

$$R_{f_{11}} = \{2\}, \qquad F_{f_{11}} = \{1\}.$$

The local factorization is thereby identified with the canonical factorization through the real coordinate support, and Variant 11 has the unique complete unit reduction procedure

$$(\{1,2\}, f_{11}) \longrightarrow_1 (\{2\}, (f_{11})_{R_{f_{11}}})$$

of length 1.

Part II provides technical preliminaries for the general theory of real support. It introduces coordinate flips on Boolean cubes and the truth-table/flip criterion for real and fictitious coordinates. It also gives a Boolean-cube formulation and direct flip-based proof of the least-coordinate-support property established generally in Paper IV. These tools are not presented as consequences of Variant 11 alone, but as the technical language required for the later construction of function spaces and the subsequent formulation of higher variants.

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Keywords: Variant 11, real support, real variable, fictitious variable, coordinate flip, truth-table criterion, least coordinate support, unit reduction.

Methodological division. The paper is divided into two parts with different functions:

Part I isolates the direct structural consequences of Variant 11

and places them within the reduction framework of Papers III–V.

Part II provides technical preliminaries for the general theory of real support.

This distinction is essential. The real/fictitious split and the local factorization are direct consequences of the Boolean function $f_{11}(p, q) = \neg q$. The canonical reduction, reduction states, and complete unit reduction procedure are their realizations within the formal framework already established in Volume I. Part II is not derived from Variant 11 alone; it prepares the general tools needed after Variant 11.

Part I
Variant 11 and Its Necessary Consequences

1 Introduction

The purpose of Part I is to study one object only: Variant 11.

Variant 11 is the Boolean function

$$f_{11} : \{0, 1\}^2 \to \{0, 1\}$$

defined by

$$f_{11}(p, q) = \neg q.$$

Throughout Part I, we write

$$f := f_{11}$$

when no ambiguity can arise.

The central point of Part I is that this function separates syntactic occurrence from real determination. The function is presented as a function of two variables,

$$f_{11}(p, q),$$

so both p and q occur syntactically. Nevertheless, only one of them determines the value of the function.

A complete local analysis of Variant 11 as a binary Boolean function has been given separately, including its formal-occurrence structure, its immediate binary environment, its

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local factorization, and its geometric reading on the Boolean square; see [1]. The present paper retains only the local structural core required by the architecture of Volume I and places it within the canonical and unit-procedural reduction framework developed in Papers III–V.

Part I first proves that

$$\mathrm{Real}(f) = \{q\}, \qquad \mathrm{Fict}(f) = \{p\}.$$

It then recalls the local factorization

$$f = g \circ \pi_q, \qquad \pi_q(p, q) = q, \qquad g(q) = \neg q,$$

and places it within the canonical-reduction language of Volume I.

For the coordinate formulation used in canonical and procedural reduction, we set

$$N_{11} := \{1, 2\}, \qquad x_1 = p, \qquad x_2 = q.$$

The corresponding coordinate supports are

$$R_f = \{2\}, \qquad F_f = \{1\}.$$

The canonical factorization is therefore

$$f = f_{R_f} \circ \pi_{R_f},$$

and the unique complete unit reduction procedure is

$$(\{1, 2\}, f) \longrightarrow_1 (\{2\}, f_{R_f}),$$

with

$$\Gamma_{\mathrm{unit}}(f) = \{\gamma\}, \qquad \ell_{\min}(f) = 1.$$

Part I does not develop a general theory of procedures, procedure spaces, trajectories, or dynamics. It distinguishes the direct structural consequences of Variant 11 from their canonical and procedural realization within the framework already established in Volume I.

2 Definition of Variant 11

Let

$$p, q \in \{0, 1\}.$$

Definition 2.1 (Variant 11)**.** Variant 11 is the Boolean function

$$f_{11} : \{0, 1\}^2 \to \{0, 1\}$$

defined by

$$\boxed{f_{11}(p, q) = \neg q.}$$

Throughout Part I, we use the abbreviation

$$f := f_{11}.$$

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We fix the following order of inputs:

$$(0,0),(0,1),(1,0),(1,1).$$

With respect to this order, the truth-vector of Variant 11 is

$$[1,0,1,0].$$

The truth table of f is:

p	q	$f(p,q)$
0	0	1
0	1	0
1	0	1
1	1	0

Remark 2.2 (Variables, Not Function Values). The values 0 and 1 are not interpreted as real or fictitious. The real/fictitious distinction applies to variables relative to the function f, not to values of f.

3 Formal Well-Definedness of Variant 11

Variant 11 is a well-defined Boolean function.

Indeed, for every pair

$$(p,q) \in \{0,1\}^2,$$

the value of

$$f(p,q) = \neg q$$

is uniquely determined.

Explicitly:

$$f(0,0) = 1,$$
$$f(0,1) = 0,$$
$$f(1,0) = 1,$$
$$f(1,1) = 0.$$

Thus f is a legitimate mathematical object:

$$\boxed{\text{Variant 11 is a well-defined Boolean function.}}$$

This is the first level of its mathematical truth.

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4 Syntactic Occurrence and Real Determination

The function is written as

$$f(p, q).$$

Thus two variables occur syntactically:

$$p \qquad \text{and} \qquad q.$$

However, syntactic occurrence is not the same as real determination.

A variable may appear in the notation of the function without influencing the value of the function. Variant 11 is a canonical minimal case in which this separation becomes visible.

Part I will show that p is syntactically present but fictitious, whereas q is real.

$$\boxed{\text{syntactic occurrence does not imply real determination.}}$$

5 Fictitiousness of p

We now prove that p is fictitious relative to f.

For a Boolean function of two variables, the variable p is fictitious relative to f precisely when

$$p \in \mathrm{Fict}(f) \quad \Longleftrightarrow \quad \forall q \in \{0,1\},\ f(0,q) = f(1,q).$$

Thus, to prove that p is fictitious, it is enough to check that changing p, while keeping q fixed, never changes the value of the function.

For every

$$q \in \{0,1\},$$

we have

$$f(0,q) = \neg q$$

and

$$f(1,q) = \neg q.$$

Therefore

$$f(0,q) = f(1,q) \qquad \text{for every } q \in \{0,1\}.$$

Changing the value of p, while keeping q fixed, never changes the value of f.

Hence

$$\boxed{p \in \mathrm{Fict}(f).}$$

The variable p occurs syntactically in $f(p,q)$, but it does not determine the value of f:

$$\boxed{p \text{ occurs syntactically but does not determine } f.}$$

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6 Realness of q

We now prove that q is real relative to f.

For a Boolean function of two variables, the variable q is real relative to f precisely when changing q can change the value of the function:

$$q \in \mathrm{Real}(f) \quad \Longleftrightarrow \quad \exists p \in \{0,1\},\ f(p,0) \neq f(p,1).$$

Thus, to prove that q is real, it is enough to show that there is at least one fixed value of p for which changing q changes the value of the function.

For every

$$p \in \{0,1\},$$

we have

$$f(p,0) = 1$$

and

$$f(p,1) = 0.$$

Thus

$$f(p,0) \neq f(p,1) \qquad \text{for every } p \in \{0,1\}.$$

Changing the value of q, while keeping p fixed, changes the value of f.

Hence

$$\boxed{q \in \mathrm{Real}(f).}$$

The variable q really determines the value of f:

$$\boxed{q \text{ really determines } f.}$$

7 The Real/Fictitious Split of Variant 11

The previous two sections show:

$$p \in \mathrm{Fict}(f),$$

and

$$q \in \mathrm{Real}(f).$$

Since the only variables of Variant 11 are p and q, this gives the full split.

Theorem 7.1 (Real/Fictitious Split of Variant 11)**.** *For*

$$f(p,q) = \neg q$$

one has

$$\boxed{\mathrm{Real}(f) = \{q\}, \qquad \mathrm{Fict}(f) = \{p\}.}$$

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Proof. The equality

$$f(0, q) = f(1, q)$$

for every $q \in \{0, 1\}$ shows that changing p never changes the value of f. Hence

$$p \in \mathrm{Fict}(f).$$

The inequality

$$f(p, 0) \neq f(p, 1)$$

for every $p \in \{0, 1\}$ shows that changing q changes the value of f. Hence

$$q \in \mathrm{Real}(f).$$

Since the only variables under consideration are p and q, the claim follows:

$$\mathrm{Real}(f) = \{q\}, \qquad \mathrm{Fict}(f) = \{p\}.$$

□

This is the central necessary consequence of Variant 11.

8 Minimality of the Split and Real Dimension

Variant 11 contains exactly one real variable and exactly one fictitious variable:

$$|\,\mathrm{Real}(f)| = 1,$$

$$|\,\mathrm{Fict}(f)| = 1.$$

Within the fixed binary presentation adopted here, Variant 11 is the minimal binary structure in which one syntactically present variable is real and one syntactically present variable is fictitious, so that formal occurrence and real determination visibly separate.

The syntactic arity of Variant 11 is

$$\mathrm{arity}(f) = 2,$$

because the function is presented as a function of two variables:

$$f(p, q).$$

However, its real dimension of determination is

$$|\,\mathrm{Real}(f)| = 1.$$

In the notation of Paper III,

$$\mathrm{PE}(f) = 1.$$

Thus Variant 11 is syntactically binary but determinationally unary: it has two syntactic variables, but only one real variable of determination.

This is the local dimensional content of the split. No gradient-depth notion and no general theory of dimensions is introduced here.

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9 Canonical Reduction of Variant 11

The autonomous local analysis of Variant 11 yields the factorization

$$f = g \circ \pi_q,$$

where

$$\pi_q : \{0,1\}^2 \to \{0,1\}, \qquad \pi_q(p,q) = q,$$

and

$$g : \{0,1\} \to \{0,1\}, \qquad g(q) = \neg q.$$

We now place this local factorization within the canonical-reduction language established in Papers III and IV. Recall that

$$N_{11} = \{1,2\}, \qquad x_1 = p, \qquad x_2 = q.$$

Since

$$\mathrm{Real}(f) = \{q\},$$

the corresponding real coordinate support is

$$R_f = \{2\}.$$

The canonical projection is

$$\pi_{R_f} : \{0,1\}^2 \to X_{R_f}.$$

Under the canonical identification

$$X_{R_f} \cong \{0,1\},$$

this projection is precisely the local projection π_q:

$$\pi_{R_f}(p,q) = q.$$

Under the same identification, the local unary function g is the canonical reduced function:

$$g = f_{R_f}, \qquad f_{R_f}(q) = \neg q.$$

For every

$$(p,q) \in \{0,1\}^2$$

we have

$$f_{R_f}(\pi_{R_f}(p,q)) = f_{R_f}(q) = \neg q = f(p,q).$$

Therefore

$$\boxed{f = f_{R_f} \circ \pi_{R_f}.}$$

Thus the local factorization

$$f = g \circ \pi_q$$

is exactly the Variant-11 instance of the canonical factorization

$$f = f_{R_f} \circ \pi_{R_f}.$$

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Proposition 9.1 (Canonical Reduction of Variant 11)**.** *Let*

$$f := f_{11}, \qquad f(p,q) = \neg q.$$

Under the coordinate identification

$$x_1 = p, \qquad x_2 = q,$$

one has

$$R_f = \{2\}.$$

Moreover,

$$f = f_{R_f} \circ \pi_{R_f},$$

where, under the canonical identification

$$X_{R_f} \cong \{0,1\},$$

the projection π_{R_f} is the local projection

$$\pi_q(p,q) = q,$$

and

$$f_{R_f}(q) = \neg q.$$

Proof. For every $(p,q) \in \{0,1\}^2$,

$$f_{R_f}(\pi_{R_f}(p,q)) = f_{R_f}(q) = \neg q = f(p,q).$$

Hence

$$f = f_{R_f} \circ \pi_{R_f}.$$

□

Thus Variant 11 is not merely the unary negation $\neg q$. It is a two-variable syntactic object whose real determination factors through the unary reduced core on q. The local projection π_q is not replaced by the canonical projection; it is recognized as its Variant-11 realization.

10 Least Coordinate Support of Variant 11

Corollary 10.1 (Least Coordinate Support of Variant 11)**.** *For Variant 11,*

$$R_f = \{2\}$$

is the least coordinate support through which f factors.

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Proof. The coordinate support $\{2\}$ is sufficient because its associated variable is q, and

$$f(p, q) = \neg q.$$

It is also necessary, since

$$f(p, 0) = 1 \qquad \text{and} \qquad f(p, 1) = 0,$$

and therefore

$$f(p, 0) \neq f(p, 1).$$

Thus the second coordinate cannot be removed. Equivalently, the real variable q cannot be removed without loss of determination. □

No general minimal support theorem is introduced in Part I.
This is only the local support consequence of Variant 11.

11 Meaning of the Reduction

The reduction of Variant 11 removes only the fictitious variable:

$$p.$$

It preserves the real variable:

$$q.$$

In the coordinate representation, this means removing coordinate 1 and preserving coordinate 2.

Thus Variant 11 shows that the fictitious variable can be removed without loss of determination:

$$\boxed{\text{the fictitious variable can be removed without loss of determination.}}$$

It also shows that the real variable cannot be removed without loss of determination:

$$\boxed{\text{the real variable cannot be removed without loss of determination.}}$$

The reduction has the form

$$f(p, q) = \neg q \quad \longrightarrow \quad f_{R_f}(q) = \neg q.$$

It removes syntactic excess and preserves real determination.

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12 Reduction States of Variant 11

In the context of Variant 11, there are exactly two relevant reduction states.

The initial state is

$$(\{1,2\}, f),$$

where coordinates 1 and 2 correspond respectively to the variables p and q, and

$$f(p,q) = \neg q.$$

The terminal state is

$$(\{2\}, f_{R_f}),$$

where the remaining coordinate 2 corresponds to the real variable q, and

$$f_{R_f}(q) = \neg q.$$

Definition 12.1 (Reduction-State Set of Variant 11)**.** The reduction-state set of Variant 11 is

$$\mathcal{R}(f) = \left\{(\{1,2\}, f), (\{2\}, f_{R_f})\right\}.$$

This is not a general reduction-state space.

It is only the minimal state structure obtained by applying the reduction framework of Paper V to Variant 11.

13 The Unit Reduction Step

Since

$$p \in \mathrm{Fict}(f),$$

the variable p may be removed without loss of determination. Under the coordinate identification $x_1 = p$, this is equivalent to

$$1 \in F_f.$$

We denote the corresponding unit reduction step by

$$\longrightarrow_1 .$$

Definition 13.1 (Unit Reduction Step of Variant 11)**.** The unit reduction step of Variant 11 is the transition

$$\boxed{(\{1,2\}, f) \longrightarrow_1 (\{2\}, f_{R_f})}$$

which removes coordinate 1, corresponding to the fictitious variable p, preserves coordinate 2, corresponding to the real variable q, and reaches the reduced function f_{R_f}.

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The step is admissible because

$$1 \in F_f,$$

equivalently because

$$p \in \mathrm{Fict}(f).$$

It preserves determination because

$$f = f_{R_f} \circ \pi_{R_f}.$$

Applying the unit-procedural language established in Paper V to Variant 11 therefore yields the unique admissible unit reduction step

$$(\{1,2\}, f) \longrightarrow_1 (\{2\}, f_{R_f}).$$

14 The Unit Procedure of Variant 11

The unit step is the unique complete unit reduction procedure of Variant 11.

Definition 14.1 (Complete Unit Reduction Procedure of Variant 11)**.** A complete unit reduction procedure of Variant 11 is a finite sequence of admissible unit steps leading from the initial state

$$(\{1,2\}, f)$$

to the terminal state

$$(\{2\}, f_{R_f}).$$

Since

$$\mathrm{Fict}(f) = \{p\},$$

equivalently

$$F_f = \{1\},$$

there is only one such procedure.

We denote it by

$$\gamma.$$

It has the form

$$\boxed{\gamma : (\{1,2\}, f) \longrightarrow_1 (\{2\}, f_{R_f}).}$$

Thus the set of complete unit reduction procedures of Variant 11 is

$$\boxed{\Gamma_{\mathrm{unit}}(f) = \{\gamma\}.}$$

Consequently,

$$\boxed{|\Gamma_{\mathrm{unit}}(f)| = 1.}$$

Variant 11 gives one complete unit reduction procedure, not a nontrivial procedure space.

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15 Length and Minimal Length

Definition 15.1 (Length of a Complete Unit Reduction Procedure)**.** The length of a complete unit reduction procedure is the number of its steps.

For

$$\gamma : (\{1,2\}, f) \longrightarrow_1 (\{2\}, f_{R_f}),$$

there is exactly one step.

Hence

$$\ell(\gamma) = 1.$$

Definition 15.2 (Minimal Unit Length of Variant 11)**.** The minimal unit length of Variant 11 is

$$\ell_{\min}(f) := \min \{\ell(P) \mid P \in \Gamma_{\text{unit}}(f)\}.$$

Proposition 15.3 (Unique Complete Unit Reduction Procedure and Minimal Length)**.** *Variant 11 has exactly one complete unit reduction procedure,*

$$\Gamma_{\text{unit}}(f) = \{\gamma\},$$

where

$$\gamma : (\{1,2\}, f) \longrightarrow_1 (\{2\}, f_{R_f}).$$

Moreover,

$$\ell(\gamma) = 1$$

and

$$\ell_{\min}(f) = 1.$$

Proof. Since

$$F_f = \{1\},$$

the only complete unit reduction procedure removes coordinate 1, corresponding to the fictitious variable p, and reaches

$$(\{2\}, f_{R_f}).$$

Hence

$$\Gamma_{\text{unit}}(f) = \{\gamma\}.$$

The procedure γ has exactly one step, so

$$\ell(\gamma) = 1.$$

Therefore the minimum over all complete unit reduction procedures is

$$\ell_{\min}(f) = 1.$$

□

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16 Necessary Consequences of Variant 11

For

$$f = f_{11}, \qquad f(p,q) = \neg q,$$

the conclusions of Part I fall into two groups.

Direct structural consequences.

$$[1,0,1,0] \text{ is the truth-vector of } f.$$

$$\boxed{\mathrm{Real}(f) = \{q\}}$$

$$\boxed{\mathrm{Fict}(f) = \{p\}}$$

$$\mathrm{arity}(f) = 2$$

$$\mathrm{PE}(f) = |\,\mathrm{Real}(f)| = 1$$

$$\boxed{f = g \circ \pi_q}$$

$$\boxed{\pi_q(p,q) = q, \qquad g(q) = \neg q}$$

Canonical and unit-procedural realizations within Volume I.

$$\boxed{N_{11} = \{1,2\}, \qquad x_1 = p, \qquad x_2 = q}$$

$$\boxed{R_f = \{2\}, \qquad F_f = \{1\}}$$

$$\boxed{\pi_{R_f}(p,q) = q}$$

$$\boxed{f_{R_f}(q) = \neg q}$$

$$\boxed{f = f_{R_f} \circ \pi_{R_f}}$$

$$\boxed{R_f = \{2\} \text{ is the least coordinate support of Variant 11}}$$

$$\boxed{\mathcal{R}(f) = \Big\{(\{1,2\}, f), (\{2\}, f_{R_f})\Big\}}$$

$$\boxed{(\{1,2\}, f) \longrightarrow_1 (\{2\}, f_{R_f})}$$

$$\boxed{\Gamma_{\mathrm{unit}}(f) = \{\gamma\}}$$

$$\boxed{|\Gamma_{\mathrm{unit}}(f)| = 1}$$

$$\boxed{\ell(\gamma) = 1}$$

$$\boxed{\ell_{\min}(f) = 1}$$

The first group consists of direct structural consequences of the Boolean function $f_{11}(p,q) = \neg q$. The second group records their canonical and unit-procedural realization within the framework established in Papers III–V.

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17 Scope of Part I

Part I stops at the local real/fictitious split of Variant 11 and its canonical and unit-procedural realization within the framework of Papers III–V. No broader procedural or dynamical structure is inferred from Variant 11 itself.

Part II
Technical Preliminaries for Real Support

18 Role of Part II

Part II has a different function from Part I.

Part I isolated the direct structural core of Variant 11,

$$f_{11}(p, q) = \neg q,$$

and placed it within the canonical and unit-procedural reduction framework of Papers III–V.

The local analysis distinguishes constancy under variation of p from value change under variation of q. We now place this local test in a general coordinate language by means of Boolean flips.

Part II is not derived from Variant 11 alone. It provides the technical language needed for the general theory of real support.

In particular, Part II introduces:

coordinate flips,

real and fictitious coordinates via flips,

the truth-table/flip criterion,

dependence on a set of coordinates,

and

the Boolean-cube form of the least coordinate support theorem.

The general least-coordinate-support result was established in Paper IV through coordinate factorization. Part II does not reintroduce that result as a new general theorem; it provides its Boolean-cube formulation and a direct proof in the coordinate-flip language developed here.

These tools prepare the next stage of the theory, where one will need a common language for comparing the real supports of different Boolean functions.

Part II provides technical preliminaries for the general theory of real support.

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19 Coordinate Flip

Let

$$x = (x_1, \ldots, x_n) \in \{0,1\}^n$$

and let

$$1 \le i \le n.$$

Definition 19.1 (Coordinate Flip)**.** The coordinate flip of x in the i-th coordinate is the point

$$x^{(i)} := (x_1, \ldots, x_{i-1}, 1 - x_i, x_{i+1}, \ldots, x_n).$$

Thus $x^{(i)}$ is obtained from x by changing exactly the i-th coordinate and leaving all other coordinates unchanged.

We denote the corresponding coordinate-flip map by

$$T_i : \{0,1\}^n \to \{0,1\}^n, \qquad T_i(x) := x^{(i)}.$$

20 Involutivity of the Coordinate Flip

The coordinate flip is involutive.

Lemma 20.1 (Involutivity of the coordinate flip)**.** *For every $x \in \{0,1\}^n$ and every $1 \le i \le n$,*

$$\boxed{\left(x^{(i)}\right)^{(i)} = x.}$$

Proof. The first flip changes x_i into $1 - x_i$. The second flip changes it back:

$$1 - (1 - x_i) = x_i.$$

All other coordinates remain unchanged in both flips. Hence

$$\left(x^{(i)}\right)^{(i)} = x.$$

□

This property is involutivity:

$$T_i^2 = \mathrm{id}\,.$$

It is not idempotence. Idempotence would mean $T_i^2 = T_i$, which is not the case for a nontrivial coordinate flip.

By definition, if $j \neq i$, then

$$x_j^{(i)} = x_j.$$

Thus a coordinate flip changes exactly one coordinate.

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21 Real and Fictitious Coordinates by Flip

Let

$$f : \{0,1\}^n \to \{0,1\}.$$

Definition 21.1 (Real Coordinate on the Boolean Cube)**.** A coordinate $i \in \{1,\dots,n\}$ is real relative to f if there exists

$$x \in \{0,1\}^n$$

such that

$$f(x) \neq f(x^{(i)}).$$

Equivalently, changing the i-th coordinate can change the value of the function.

Definition 21.2 (Fictitious Coordinate on the Boolean Cube)**.** A coordinate $i \in \{1,\dots,n\}$ is fictitious relative to f if for every

$$x \in \{0,1\}^n$$

one has

$$f(x) = f(x^{(i)}).$$

Equivalently, changing the i-th coordinate never changes the value of the function.

We define the real and fictitious coordinate supports by

$$R_f := \{i \in \{1,\dots,n\} : i \text{ is real relative to } f\},$$

and

$$F_f := \{1,\dots,n\} \setminus R_f.$$

If the formal variables are denoted by $x_1,\dots,x_n$, then the corresponding variable sets are

$$\mathrm{Real}(f) = \{x_i : i \in R_f\},$$

and

$$\mathrm{Fict}(f) = \{x_i : i \in F_f\}.$$

Thus the flip criterion identifies real and fictitious coordinates; it does not replace the variable-level distinction established in Paper III.

22 Truth-Table / Flip Criterion

The preceding definitions yield the following criterion.

Proposition 22.1 (Truth-Table/Flip Criterion)**.** *Let*

$$f : \{0,1\}^n \to \{0,1\}$$

and let

$$1 \leq i \leq n.$$

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Then

$$\boxed{i \in R_f \quad \Longleftrightarrow \quad \exists x \in \{0,1\}^n : \ f(x) \neq f(x^{(i)}).}$$

Moreover,

$$\boxed{i \in F_f \quad \Longleftrightarrow \quad \forall x \in \{0,1\}^n : \ f(x) = f(x^{(i)}).}$$

Proof. Both equivalences follow immediately from the definitions of real and fictitious coordinates via the coordinate flip. □

This criterion is the truth-table test for whether a coordinate is real or fictitious: one checks whether flipping that coordinate can change the value of the function.

23 Dependence on a Set of Coordinates

Let

$$S \subseteq \{1, \ldots, n\}.$$

Definition 23.1 (Dependence on a set of coordinates)**.** A function

$$f : \{0,1\}^n \to \{0,1\}$$

depends only on the coordinates in S if for all

$$x, y \in \{0,1\}^n$$

such that

$$x_j = y_j \qquad \text{for all } j \in S,$$

one has

$$f(x) = f(y).$$

In other words, once the coordinates in S are fixed, the value of f is fixed. Coordinates outside S do not affect the value of f.

24 Least Coordinate Support on the Boolean Cube

Paper IV established generally that the real support is the least coordinate support through which a function factors. We now formulate this property on the Boolean cube and give a direct proof using the coordinate-flip language developed in Part II.

Theorem 24.1 (Least Coordinate Support on the Boolean Cube)**.** *Let*

$$f : \{0,1\}^n \to \{0,1\}.$$

Then:

1. *f depends only on R_f;*

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2. if $S \subseteq \{1, \dots, n\}$ is such that f depends only on S, then

$$R_f \subseteq S.$$

Consequently,

$$\boxed{R_f \text{ is the least coordinate support of } f.}$$

Proof. We first prove that f depends only on R_f.

Let

$$x, y \in \{0,1\}^n$$

agree on all coordinates in R_f. Then x and y may differ only on coordinates in

$$F_f.$$

Let

$$i_1, \dots, i_m$$

be the finite list of coordinates on which x and y differ. Each i_r belongs to F_f.

Construct a finite chain

$$x = z^{(0)}, z^{(1)}, \dots, z^{(m)} = y$$

such that $z^{(r)}$ is obtained from $z^{(r-1)}$ by flipping the coordinate i_r.

Since

$$i_r \in F_f,$$

we have

$$f(z^{(r-1)}) = f(z^{(r)})$$

for every $r = 1, \dots, m$.

Therefore

$$f(x) = f(z^{(0)}) = f(z^{(1)}) = \cdots = f(z^{(m)}) = f(y).$$

Thus f depends only on R_f.

Now suppose that f depends only on a set

$$S \subseteq \{1, \dots, n\}.$$

Let

$$i \in R_f.$$

By the truth-table/flip criterion, there exists

$$x \in \{0,1\}^n$$

such that

$$f(x) \neq f(x^{(i)}).$$

If $i \notin S$, then x and $x^{(i)}$ agree on all coordinates in S, because they differ only in coordinate i. Since f depends only on S, this would imply

$$f(x) = f(x^{(i)}),$$

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contradicting the choice of x.

Therefore

$$i \in S.$$

Since $i \in R_f$ was arbitrary, we have

$$R_f \subseteq S.$$

Hence R_f is contained in every coordinate set on which f depends. Since f itself depends only on R_f, it follows that R_f is the least coordinate support of f. □

Remark 24.2. The theorem proves a least-support property, not merely a minimal-support property. The set R_f is contained in every coordinate support of f.

25 Relation of Part II to Part I

For Variant 11,

$$f = f_{11}, \qquad f(p, q) = \neg q,$$

Part I showed directly that

$$\mathrm{Real}(f) = \{q\}, \qquad \mathrm{Fict}(f) = \{p\}.$$

Under the identification

$$x_1 = p, \qquad x_2 = q,$$

Part I then obtained

$$R_f = \{2\}, \qquad F_f = \{1\},$$

and showed that $R_f = \{2\}$ is the least coordinate support of Variant 11.

Part II provides the Boolean technical language for testing real and fictitious coordinates and for expressing the least-support property for arbitrary Boolean functions:

$$f : \{0, 1\}^n \to \{0, 1\}.$$

Thus Part II does not replace the direct analysis of Variant 11. It places the local constancy/variation test in a general flip language and prepares the tools needed for later work, where one must compare real supports across different functions.

In this sense:

Part I gives the local split and its canonical realization.

Part II gives the technical support language.

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26 Scope of the Whole Paper

The whole paper has two principal parts, while Part I itself contains two carefully distinguished levels.

First, it retains the direct structural core of Variant 11 required by Volume I:

$$f_{11}(p,q) = \neg q,$$

$$\mathrm{Real}(f_{11}) = \{q\}, \qquad \mathrm{Fict}(f_{11}) = \{p\},$$

together with its local factorization

$$f_{11} = g \circ \pi_q.$$

Second, it places this local structure within the canonical and unit-procedural reduction framework of Papers III–V:

$$R_{f_{11}} = \{2\}, \qquad F_{f_{11}} = \{1\},$$

$$f_{11} = (f_{11})_{R_{f_{11}}} \circ \pi_{R_{f_{11}}},$$

and

$$(\{1,2\}, f_{11}) \longrightarrow_1 (\{2\}, (f_{11})_{R_{f_{11}}}).$$

Third, Part II introduces the Boolean technical tools required for the later theory of real support:

$$x^{(i)},$$

$$R_f, \qquad F_f,$$

truth-table / flip criterion,

and

the Boolean-cube realization of R_f as the least coordinate support of f.

Part II stops at the Boolean support tools required for the subsequent comparison of real supports across functions.

27 Conclusion

This paper established two things.

First, Part I retained the direct structural core of Variant 11,

$$f = f_{11}, \qquad f_{11}(p,q) = \neg q,$$

for which

$$\mathrm{Real}(f_{11}) = \{q\}, \qquad \mathrm{Fict}(f_{11}) = \{p\}.$$

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This separation is witnessed explicitly by the truth table: changing p never changes the value of the function, whereas changing q does. Variant 11 therefore has syntactic arity 2, but only one real dimension of determination:

$$\mathrm{PE}(f_{11}) = 1.$$

The autonomous local analysis records the factorization

$$f_{11} = g \circ \pi_q, \qquad \pi_q(p, q) = q, \qquad g(q) = \neg q.$$

The present paper places this local structure within the canonical language of Volume I. Under

$$x_1 = p, \qquad x_2 = q,$$

one has

$$R_{f_{11}} = \{2\}, \qquad F_{f_{11}} = \{1\},$$

and

$$f_{11} = (f_{11})_{R_{f_{11}}} \circ \pi_{R_{f_{11}}}.$$

The local projection π_q is thereby recognized, under the canonical identification of the reduced domain, as the Variant-11 realization of $\pi_{R_{f_{11}}}$.

Applying the unit-procedural framework of Paper V gives

$$(\{1, 2\}, f_{11}) \longrightarrow_1 (\{2\}, (f_{11})_{R_{f_{11}}}),$$

and the unique complete unit reduction procedure:

$$\Gamma_{\mathrm{unit}}(f_{11}) = \{\gamma\}, \qquad \ell_{\min}(f_{11}) = 1.$$

The local factorization $f_{11} = g \circ \pi_q$ belongs to the autonomous structural analysis of Variant 11. Its expression as a canonical real-support factorization and as a complete unit reduction procedure is the additional genealogical contribution made here within Volume I.

Second, Part II introduced the Boolean technical language needed for the subsequent theory of real support:

$$x^{(i)},$$

$$i \in R_f \iff \exists x : \ f(x) \neq f(x^{(i)}),$$

$$i \in F_f \iff \forall x : \ f(x) = f(x^{(i)}),$$

The Boolean-cube form of the least-coordinate-support property established generally in Paper IV.

The paper therefore has a double role:

Variant 11 provides the canonical minimal cell of visible separation.

Here the visible separation is the separation between formal occurrence and real determination.

The technical preliminaries establish the language of real support.

These two components complete the role assigned to Paper VI within the present volume.

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Paper VII

The Space of Epistemic Cores and Its Layers. From Syntactic Functions to Real Support

Piotr Dariusz Wójcik

Version 1.3

Abstract

This paper constructs the space in which later variants of the theory can be formulated. It begins with finite coordinate sets $N \in [\mathbb{N}]^{<\omega}$, local spaces of syntactic functions

$$\mathcal{F}_N = \{f : \{0,1\}^N \to \{0,1\}\},$$

and their disjoint union

$$\mathcal{F}_{\text{syn}} = \bigsqcup_{N \in [\mathbb{N}]^{<\omega}} \mathcal{F}_N.$$

For a syntactic function f, its real coordinate support is

$$R_f = \{i \in N : x_i \in \text{Real}(f)\} \subseteq N.$$

Its fictitious coordinate support is

$$F_f := N \setminus R_f.$$

The canonical reduction of f is its epistemic core

$$f_\infty := f_{R_f} : \{0,1\}^{R_f} \to \{0,1\},$$

satisfying

$$f = f_\infty \circ \pi^N_{R_f}.$$

This defines an idempotent reduction operator

$$\rho : \mathcal{F}_{\text{syn}} \to \mathcal{F}_{\text{syn}}, \qquad \rho(f) = f_\infty.$$

The main space of the paper is

$$X_\infty := \text{Im}(\rho),$$

the global space of finite epistemic cores. Equivalently,

$$X_\infty = \bigsqcup_{K \in [\mathbb{N}]^{<\omega}} \left\{ g : \{0,1\}^K \to \{0,1\} \;\middle|\; \text{Fict}(g) = \varnothing \right\}.$$

Equivalently, for every

$$g : \{0,1\}^K \to \{0,1\}$$

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in X_∞,

$$F_g = \varnothing, \qquad R_g = K.$$

The paper proves

$$X_\infty = \mathrm{Im}(\rho) = \mathrm{Fix}(\rho),$$

defines epistemic width

$$\mathrm{PE}_B(g) = |\,\mathrm{Real}(g)| = |R_g|,$$

and introduces the layers

$$L_d = \{g \in X_\infty : \mathrm{PE}_B(g) = d\},$$

which stratify X_∞. Finally, it records that the epistemic core of Variant 11 belongs to L_1, and it makes available the support-inclusion relation whose structural analysis is left to Variant 12.

Keywords: epistemic core, syntactic function, real support, reduction, X_∞, epistemic width, layers, support comparison, Variant 11, Variant 12.

1 Introduction

Paper VI retained the local real/fictitious split of Variant 11, placed it within canonical reduction, and established the Boolean criterion for real coordinate support. The present paper uses that language to construct the ambient space of epistemic cores.

The starting point is a finite coordinate domain

$$N \in [\mathbb{N}]^{<\omega}$$

and a syntactic function

$$f : \{0,1\}^N \to \{0,1\}.$$

Such a function may contain coordinates that occur syntactically but do not determine its value. Here the declared coordinate domain belongs to the syntactic data of the function and may contain coordinates that do not participate in real determination. The reduction to real support removes this syntactic excess and yields the epistemic core of the function.

The central movement of the paper is:

$$\boxed{\mathcal{F}_{\mathrm{syn}} \xrightarrow{\rho} X_\infty = \bigsqcup_{d \geq 0} L_d.}$$

Here $\mathcal{F}_{\mathrm{syn}}$ is the global space before reduction, ρ is the reduction to epistemic core, X_∞ is the global space of finite epistemic cores, and L_d are the layers determined by epistemic width.

The main statement is:

$$\boxed{X_\infty \text{ is the global space of finite epistemic cores.}}$$

This paper does not develop Variant 12. It constructs the space in which Variant 12 can later be formulated.

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2 Finite Coordinate Sets

Let

$$[\mathbb{N}]^{<\omega}$$

denote the set of all finite subsets of $\mathbb{N}$.

Definition 2.1 (Finite Boolean coordinate cube)**.** For

$$N \in [\mathbb{N}]^{<\omega},$$

define

$$\{0,1\}^N = \{x : N \to \{0,1\}\}.$$

If $x \in \{0,1\}^N$ and $i \in N$, we write

$$x_i := x(i).$$

If $N = \varnothing$, then

$$\{0,1\}^{\varnothing}$$

is a one-point set. We denote its unique element by $*$:

$$\boxed{\{0,1\}^{\varnothing} = \{*\}.}$$

Thus the basic domains are finite Boolean coordinate cubes

$$\{0,1\}^N, \qquad N \in [\mathbb{N}]^{<\omega}.$$

3 Local Spaces of Syntactic Functions

Definition 3.1 (Local space of syntactic functions)**.** For

$$N \in [\mathbb{N}]^{<\omega},$$

define

$$\boxed{\mathcal{F}_N = \{f : \{0,1\}^N \to \{0,1\}\}.}$$

An element $f \in \mathcal{F}_N$ is called a syntactic function on N. Such a function is not necessarily an epistemic core, because it may contain fictitious coordinates.

For Variant 11, set

$$N_{11} := \{1,2\}, \qquad x_1 = p, \qquad x_2 = q.$$

The syntactic function is

$$f_{11} : \{0,1\}^{N_{11}} \to \{0,1\}, \qquad f_{11}(p,q) = \neg q.$$

Both variables p and q occur syntactically, but only q is real relative to f_{11}.

Remark 3.2 (Variables and coordinate indices). The symbols p and q name the variables of Variant 11. Their coordinate representation is fixed by

$$x_1 = p, \qquad x_2 = q.$$

Thus $1, 2 \in \mathbb{N}$ are coordinate indices, whereas p and q are the corresponding variable names.

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4 The Global Space of Syntactic Functions

Definition 4.1 (Global space of syntactic functions)**.** Define

$$\boxed{\mathcal{F}_{\mathrm{syn}} = \bigsqcup_{N \in [\mathbb{N}]^{<\omega}} \mathcal{F}_N.}$$

The disjoint union records the coordinate domain as part of the data. Thus two functions with the same formal logical shape but defined on different coordinate sets belong to different components.

$$\boxed{\mathcal{F}_{\mathrm{syn}} \text{ is the global pre-reduction space.}}$$

5 Real Support of a Syntactic Function

Let

$$f : \{0,1\}^N \to \{0,1\}.$$

Define its real coordinate support by

$$\boxed{R_f := \{i \in N : x_i \in \mathrm{Real}(f)\}.}$$

Define its fictitious coordinate support by

$$\boxed{F_f := N \setminus R_f.}$$

The corresponding variable sets are

$$\mathrm{Real}(f) = \{x_i : i \in R_f\},$$

and

$$\mathrm{Fict}(f) = \{x_i : i \in F_f\}.$$

By the least-support theorem for real coordinates, R_f is the least coordinate support of f. Namely, f depends only on R_f, and if f depends only on $S \subseteq N$, then

$$R_f \subseteq S.$$

Thus

$$\boxed{R_f \text{ is the least coordinate support of } f.}$$

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6 The Reduction Projection

Let

$$f : \{0,1\}^N \to \{0,1\}$$

and let $R_f \subseteq N$ be its real support.

Definition 6.1 (Reduction projection)**.** The reduction projection of f is the map

$$\boxed{\pi^N_{R_f} : \{0,1\}^N \to \{0,1\}^{R_f}}$$

defined by

$$\boxed{\pi^N_{R_f}(x) = (x_i)_{i \in R_f}.}$$

When N is clear, we write simply

$$\pi_{R_f}$$

instead of

$$\pi^N_{R_f}.$$

If $R_f = \varnothing$, then

$$\pi^N_{R_f} : \{0,1\}^N \to \{0,1\}^{\varnothing}$$

is the unique map into the one-point set $\{*\}$.

7 The Epistemic Core of a Syntactic Function

Let

$$f : \{0,1\}^N \to \{0,1\}.$$

Definition 7.1 (Epistemic core)**.** The epistemic core of f is the unique function

$$\boxed{f_\infty := f_{R_f} : \{0,1\}^{R_f} \to \{0,1\}}$$

such that

$$\boxed{f = f_\infty \circ \pi^N_{R_f}.}$$

Equivalently,

$$f(x) = f_\infty(\pi^N_{R_f}(x)) \qquad \text{for all } x \in \{0,1\}^N.$$

The index ∞ indicates membership in the global space of finite epistemic cores, not an infinite input domain.

Proposition 7.2 (Existence and Uniqueness of the Epistemic Core)**.** *Every syntactic function has a unique epistemic core.*

More precisely, for every

$$f : \{0,1\}^N \to \{0,1\}$$

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there exists a unique function

$$f_\infty : \{0,1\}^{R_f} \to \{0,1\}$$

such that

$$f = f_\infty \circ \pi^N_{R_f}.$$

Proof. Since R_f is the least coordinate support of f, the value of $f(x)$ depends only on the coordinates of x indexed by R_f.

For $u \in \{0,1\}^{R_f}$, choose any $x \in \{0,1\}^N$ such that

$$\pi^N_{R_f}(x) = u.$$

Define

$$f_\infty(u) := f(x).$$

This is well-defined: if

$$\pi^N_{R_f}(x) = \pi^N_{R_f}(y),$$

then x and y agree on R_f, hence

$$f(x) = f(y).$$

Thus

$$f = f_\infty \circ \pi^N_{R_f}.$$

For uniqueness, suppose

$$g : \{0,1\}^{R_f} \to \{0,1\}$$

also satisfies

$$f = g \circ \pi^N_{R_f}.$$

For $u \in \{0,1\}^{R_f}$, choose $x \in \{0,1\}^N$ with

$$\pi^N_{R_f}(x) = u.$$

Then

$$g(u) = g(\pi^N_{R_f}(x)) = f(x) = f_\infty(u).$$

Hence

$$g = f_\infty.$$

□

Proposition 7.3 (The core has no fictitious coordinates)**.** *For every syntactic function*

$$f : \{0,1\}^N \to \{0,1\},$$

its epistemic core

$$f_\infty : \{0,1\}^{R_f} \to \{0,1\}$$

satisfies

$$\boxed{\mathrm{Fict}(f_\infty) = \varnothing.}$$

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Equivalently, at the coordinate-support level,

$$\boxed{F_{f_\infty} = \varnothing, \qquad R_{f_\infty} = R_f.}$$

At the variable level,

$$\mathrm{Real}(f_\infty) = \{x_i : i \in R_f\}.$$

Proof. Let $i \in R_f$. Since i is real for f, there exist $x, y \in \{0,1\}^N$, differing only in coordinate i, such that

$$f(x) \neq f(y).$$

Let

$$u = \pi^N_{R_f}(x), \qquad v = \pi^N_{R_f}(y).$$

Then u and v differ only in coordinate i, and

$$f_\infty(u) = f(x), \qquad f_\infty(v) = f(y).$$

Hence

$$f_\infty(u) \neq f_\infty(v).$$

Thus every $i \in R_f$ is a real coordinate for f_∞. Therefore

$$R_{f_\infty} = R_f, \qquad F_{f_\infty} = \varnothing,$$

and hence

$$\mathrm{Fict}(f_\infty) = \varnothing.$$

□

8 Epistemic Functions

Definition 8.1 (Epistemic function)**.** Let

$$K \in [\mathbb{N}]^{<\omega}.$$

A function

$$g : \{0,1\}^K \to \{0,1\}$$

is called epistemic if

$$\boxed{\mathrm{Fict}(g) = \varnothing.}$$

Equivalently, at the coordinate-support level,

$$\boxed{g \text{ is epistemic} \quad \Longleftrightarrow \quad F_g = \varnothing \quad \Longleftrightarrow \quad R_g = K.}$$

At the variable level,

$$\mathrm{Real}(g) = \{x_i : i \in K\}.$$

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"Epistemic" names the internal property of having no fictitious variables, whereas "epistemic core" names the role of such a function as a canonical reduction output and a fixed point of ρ.

The case $K = \varnothing$ is allowed. Then

$$g : \{0,1\}^{\varnothing} \to \{0,1\}$$

is a function on a one-point domain. Such functions have no fictitious coordinates and represent constant epistemic cores.

9 The Reduction Map ρ

Definition 9.1 (Reduction map)**.** Define

$$\boxed{\rho : \mathcal{F}_{\text{syn}} \to \mathcal{F}_{\text{syn}}}$$

by

$$\boxed{\rho(f) = f_\infty.}$$

If $f \in \mathcal{F}_N$, then

$$\rho(f) = f_\infty \in \mathcal{F}_{R_f}.$$

Thus ρ may move a function from one component of the disjoint union to another:

$$\mathcal{F}_N \longrightarrow \mathcal{F}_{R_f}.$$

Proposition 9.2 (Idempotence of Reduction)**.**

$$\boxed{\rho^2 = \rho.}$$

Proof. Let $f \in \mathcal{F}_{\text{syn}}$. Then

$$\rho(f) = f_\infty.$$

Since

$$\text{Fict}(f_\infty) = \varnothing,$$

the epistemic core of f_∞ is f_∞ itself:

$$(f_\infty)_\infty = f_\infty.$$

Therefore

$$\rho^2(f) = \rho(\rho(f)) = \rho(f_\infty) = (f_\infty)_\infty = f_\infty = \rho(f).$$

Hence

$$\rho^2 = \rho.$$

□

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10 The Space X_∞

Definition 10.1 (The space of epistemic cores)**.** Define

$$\boxed{X_\infty := \mathrm{Im}(\rho).}$$

Thus

$$X_\infty = \{\rho(f) : f \in \mathcal{F}_{\mathrm{syn}}\}.$$

Equivalently,

$$\boxed{X_\infty = \bigsqcup_{K \in [\mathbb{N}]^{<\omega}} \left\{ g : \{0,1\}^K \to \{0,1\} \;\middle|\; \mathrm{Fict}(g) = \varnothing \right\}.}$$

Equivalently, for every

$$g : \{0,1\}^K \to \{0,1\}$$

in X_∞,

$$F_g = \varnothing, \qquad R_g = K.$$

Therefore

$$\boxed{X_\infty \text{ is the global space of finite epistemic cores.}}$$

11 Cores as Fixed Points of Reduction

Define

$$\mathrm{Fix}(\rho) := \{f \in \mathcal{F}_{\mathrm{syn}} : \rho(f) = f\}.$$

Theorem 11.1 (Fixed-Point Characterization of X_∞)**.**

$$\boxed{X_\infty = \mathrm{Fix}(\rho).}$$

Proof. Let $g \in X_\infty$. Then $g = \rho(f)$ for some $f \in \mathcal{F}_{\mathrm{syn}}$. Since $\rho^2 = \rho$,

$$\rho(g) = \rho(\rho(f)) = \rho(f) = g.$$

Thus $g \in \mathrm{Fix}(\rho)$.

Conversely, if $g \in \mathrm{Fix}(\rho)$, then

$$g = \rho(g) \in \mathrm{Im}(\rho) = X_\infty.$$

Hence

$$X_\infty = \mathrm{Fix}(\rho).$$

□

Thus

$$\boxed{X_\infty = \mathrm{Im}(\rho) = \mathrm{Fix}(\rho).}$$

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12 Epistemic Width

Definition 12.1 (Epistemic width)**.** For

$$g \in X_\infty,$$

define

$$\boxed{\mathrm{PE}_B(g) := |\operatorname{Real}(g)| = |R_g|.}$$

Since $g \in X_\infty$, there exists

$$K \in [\mathbb{N}]^{<\omega}$$

such that

$$g : \{0,1\}^K \to \{0,1\}$$

and

$$\operatorname{Fict}(g) = \varnothing.$$

Equivalently,

$$F_g = \varnothing, \qquad R_g = K.$$

Hence

$$\boxed{\mathrm{PE}_B(g) = |K|.}$$

If $g = f_\infty$, then

$$R_g = R_f,$$

and therefore

$$\boxed{\mathrm{PE}_B(f_\infty) = |R_f|.}$$

Recalling the notation of Paper III,

$$\boxed{\mathrm{PE}_B(f_\infty) = \mathrm{PE}(f) = |R_f|.}$$

Epistemic width measures width, not depth. No depth invariant is introduced here.

13 Epistemic Layers

Definition 13.1 (Epistemic layer)**.** For each

$$d \geq 0,$$

define

$$\boxed{L_d = \{g \in X_\infty : \mathrm{PE}_B(g) = d\}.}$$

Equivalently,

$$L_d = \{g \in X_\infty : |\operatorname{Real}(g)| = d\} = \{g \in X_\infty : |R_g| = d\}.$$

Since every $g \in X_\infty$ has no fictitious coordinates, one may also write

$$\boxed{L_d = \left\{ g : \{0,1\}^K \to \{0,1\} \;\middle|\; K \in [\mathbb{N}]^{<\omega},\ \operatorname{Fict}(g) = \varnothing,\ |K| = d \right\}.}$$

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14 Stratification of X_∞

Theorem 14.1 (Epistemic-Width Stratification of X_∞)**.** *The family*

$$\{L_d\}_{d\geq 0}$$

forms a partition of X_∞*:*

$$\boxed{X_\infty = \bigsqcup_{d\geq 0} L_d.}$$

Proof. Let $g \in X_\infty$. Then

$$g : \{0,1\}^K \to \{0,1\}$$

for some finite coordinate set K, and

$$\mathrm{Fict}(g) = \varnothing.$$

Hence

$$R_g = K$$

and

$$\mathrm{PE}_B(g) = |K|.$$

Thus $g \in L_d$ for the unique integer $d = |K|$.

If $g \in L_d \cap L_e$, then

$$\mathrm{PE}_B(g) = d = e.$$

Therefore the layers are pairwise disjoint and

$$X_\infty = \bigsqcup_{d\geq 0} L_d.$$

□

15 The Zero Layer

The zero layer is

$$L_0 = \{g \in X_\infty : \mathrm{PE}_B(g) = 0\}.$$

If $g \in L_0$, then its coordinate domain is empty:

$$g : \{0,1\}^\varnothing \to \{0,1\}.$$

Since

$$\{0,1\}^\varnothing = \{*\},$$

there are exactly two such cores:

$$g_0(*) = 0, \qquad g_1(*) = 1.$$

Thus

$$\boxed{L_0 \text{ is the layer of constant epistemic cores, and } |L_0| = 2.}$$

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16 The Unary Layer

The unary layer is

$$L_1 = \{g \in X_\infty : \mathrm{PE}_B(g) = 1\}.$$

It consists of epistemic cores with exactly one real coordinate.

$$\boxed{L_1 \text{ is the layer of unary epistemic cores.}}$$

Remark 16.1 (Unary cores). For each singleton coordinate set $\{i\}$, there are exactly two nonconstant unary epistemic cores:

$$u \longmapsto u_i, \qquad u \longmapsto 1 - u_i,$$

where

$$u \in \{0,1\}^{\{i\}}.$$

Both have real coordinate support $\{i\}$.

17 Variant 11 as an Element of L_1

For Variant 11, recall the identification

$$N_{11} = \{1,2\}, \qquad x_1 = p, \qquad x_2 = q.$$

The syntactic function is

$$f_{11} : \{0,1\}^{N_{11}} \to \{0,1\}, \qquad f_{11}(p,q) = \neg q.$$

It is a function on the two-coordinate domain N_{11}. Therefore one should not write

$$f_{11} \in L_1.$$

The correct statement concerns its epistemic core.

Corollary 17.1 (Epistemic Core of Variant 11)**.** *The epistemic core of Variant 11 belongs to the unary layer:*

$$\boxed{(f_{11})_\infty \in L_1.}$$

Proof. From Paper VI,

$$R_{f_{11}} = \{2\}, \qquad F_{f_{11}} = \{1\}.$$

Hence

$$(f_{11})_\infty = (f_{11})_{R_{f_{11}}} : \{0,1\}^{\{2\}} \to \{0,1\}$$

is given by

$$(f_{11})_\infty(u) = 1 - u_2, \qquad u \in \{0,1\}^{\{2\}}.$$

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Under the identification $x_2 = q$, this is the unary function

$$q \longmapsto \neg q.$$

At the coordinate-support level,

$$R_{(f_{11})_\infty} = \{2\}.$$

At the variable level,

$$\mathrm{Real}((f_{11})_\infty) = \{q\}.$$

Therefore

$$\mathrm{PE}_B((f_{11})_\infty) = 1.$$

By the definition of L_1, it follows that

$$(f_{11})_\infty \in L_1.$$

□

18 What Is Not Introduced Here

This paper establishes the space of epistemic cores only. It does not introduce depth invariants, procedure spaces, trajectory geometry, or global dynamics. Variant 12 and Variant 13 are not developed here.

19 Technical Preparation for Variant 12

Although Variant 12 is not developed here, the present construction makes available the support-inclusion relation

$$g \preceq_R h \quad \Longleftrightarrow \quad R_g \subseteq R_h$$

for $g, h \in X_\infty$. Thus the comparison concerns inclusion of the coordinate supports that represent real determination.

Its preorder structure, the associated equivalence relation, and the resulting quotient order are established in Variant 12. The present paper records only that support comparison is now formally available on X_∞.

the structural theory of $\preceq_R$ belongs to Variant 12.

20 Consequences

The construction of the present paper establishes:

$$\boxed{\mathcal{F}_{\mathrm{syn}} = \bigsqcup_{N \in [\mathbb{N}]^{<\omega}} \mathcal{F}_N}$$

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as the global pre-reduction space,

$$\boxed{\rho : \mathcal{F}_{\text{syn}} \to \mathcal{F}_{\text{syn}}, \qquad \rho(f) = f_\infty}$$

as the idempotent reduction,

$$\boxed{X_\infty = \text{Im}(\rho) = \text{Fix}(\rho)}$$

as the space of epistemic cores,

$$\boxed{X_\infty = \bigsqcup_{d \geq 0} L_d}$$

as the width stratification,
and

$$\boxed{(f_{11})_\infty \in L_1}$$

as the placement of Variant 11.
The support-inclusion relation

$$g \preceq_R h \quad \Longleftrightarrow \quad R_g \subseteq R_h$$

is now formally available, while its preorder structure and quotient order belong to Variant 12.

The central chain is:

$$\boxed{\mathcal{F}_{\text{syn}} \xrightarrow{\rho} X_\infty = \bigsqcup_{d \geq 0} L_d.}$$

21 Conclusion

This paper constructed the space of epistemic cores.

It began with finite coordinate sets

$$N \in [\mathbb{N}]^{<\omega}$$

and local syntactic function spaces

$$\mathcal{F}_N = \{f : \{0,1\}^N \to \{0,1\}\}.$$

These were assembled into the global pre-reduction space

$$\mathcal{F}_{\text{syn}} = \bigsqcup_{N \in [\mathbb{N}]^{<\omega}} \mathcal{F}_N.$$

For every syntactic function f, the canonical reduction produces its epistemic core

$$f_\infty = f_{R_f}.$$

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This defines the idempotent reduction

$$\rho : \mathcal{F}_{\text{syn}} \to \mathcal{F}_{\text{syn}}, \qquad \rho(f) = f_\infty,$$

with

$$\rho^2 = \rho.$$

The space

$$X_\infty$$

is defined as the image of this reduction:

$$X_\infty = \text{Im}(\rho).$$

Equivalently,

$$X_\infty = \text{Fix}(\rho).$$

Thus:

$\boxed{X_\infty \text{ is the global space of finite epistemic cores.}}$

Epistemic width is given by

$$\text{PE}_B(g) = |\,\text{Real}(g)| = |R_g|,$$

and the layers L_d stratify X_∞:

$$X_\infty = \bigsqcup_{d \geq 0} L_d.$$

The epistemic core of Variant 11 satisfies

$$(f_{11})_\infty \in L_1.$$

Finally, the support-inclusion relation on X_∞ prepares the formal ground for Variant 12, without establishing its preorder or quotient-order structure here.

$\boxed{\text{Paper VII provides the space in which Variant 12 can be formulated.}}$

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Paper VIII

The Order of Real Support From Epistemic Cores to Support Classes

Piotr Dariusz Wójcik

Version 1.3

Abstract

This paper formulates Variant 12 as the order of real support. Variant 11 separates syntactic occurrence from real determination; Variant 12 establishes the first order structure after this separation. The starting point is the space X_∞ of finite epistemic cores. For $g, h \in X_\infty$, let R_g and R_h denote the coordinate supports representing the real-variable sets $\mathrm{Real}(g)$ and $\mathrm{Real}(h)$. The support preorder is defined by

$$g \preceq_R h \quad \Longleftrightarrow \quad R_g \subseteq R_h.$$

The paper proves that $\preceq_R$ is a preorder on X_∞. Since this relation need not be antisymmetric on X_∞, the paper introduces the equivalence relation

$$g \sim_R h \quad \Longleftrightarrow \quad R_g = R_h,$$

the quotient space of support classes

$$\mathcal{S}_R := X_\infty / \sim_R,$$

and the induced relation

$$[g]_R \leq_R [h]_R \quad \Longleftrightarrow \quad R_g \subseteq R_h.$$

This induced relation is proved to be a partial order on $\mathcal{S}_R$. The paper also gives the external representation

$$\Phi([g]_R) = R_g$$

of support classes, without reducing the meaning of Variant 12 to set inclusion. Finally, it records compatibility with epistemic width and with the width layers L_d. Thus Variant 12 establishes the order of real support.

Keywords: Variant 12, real support, epistemic core, support preorder, support class, quotient order, epistemic width.

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1 Introduction: Variant 12 after Variant 11

Let

$$X_\infty$$

denote the space of finite epistemic cores. For every

$$g \in X_\infty,$$

let

$$R_g = \{i : x_i \in \mathrm{Real}(g)\}$$

denote its real coordinate support. The set

$$\mathrm{Real}(g)$$

records the real variables of the core, whereas R_g is their coordinate representation.

Variant 11 separates syntactic occurrence from real determination. Paper VII made the support-inclusion comparison available on X_∞. Variant 12 establishes the first order structure after this separation: epistemic cores are compared by inclusion of their real coordinate supports.

The central relation is

$$g \preceq_R h \quad \Longleftrightarrow \quad R_g \subseteq R_h.$$

The main statement of this paper is:

$$\boxed{\text{Variant 12 establishes the order of real support.}}$$

The paper establishes order, not composition.

2 Support Preorder

Definition 2.1 (Support Preorder)**.** Let

$$g, h \in X_\infty.$$

Define

$$\boxed{g \preceq_R h \quad \Longleftrightarrow \quad R_g \subseteq R_h.}$$

The relation $\preceq_R$ compares epistemic cores by inclusion of real coordinate supports. Thus every real coordinate of g is preserved within the real support of h. Equivalently, the real-variable content represented by g is contained in that represented by h.

Proposition 2.2 (Support Preorder)**.** *The relation*

$$\preceq_R$$

is a preorder on X_∞.

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Proof. We prove reflexivity and transitivity.

Let

$$g \in X_\infty.$$

Since every set is contained in itself,

$$R_g \subseteq R_g.$$

Hence

$$g \preceq_R g.$$

Thus $\preceq_R$ is reflexive.

Now let

$$g, h, k \in X_\infty$$

and suppose that

$$g \preceq_R h \qquad \text{and} \qquad h \preceq_R k.$$

By definition,

$$R_g \subseteq R_h$$

and

$$R_h \subseteq R_k.$$

Therefore

$$R_g \subseteq R_k,$$

and so

$$g \preceq_R k.$$

Thus $\preceq_R$ is transitive. □

The relation $\preceq_R$ need not be antisymmetric on X_∞. Fix

$$i \in \mathbb{N}$$

and define two unary epistemic cores

$$g_i^+, g_i^- : \{0,1\}^{\{i\}} \to \{0,1\}$$

by

$$g_i^+(u) = u_i$$

and

$$g_i^-(u) = 1 - u_i.$$

Then

$$g_i^+ \neq g_i^-,$$

but

$$R_{g_i^+} = R_{g_i^-} = \{i\}.$$

Hence

$$g_i^+ \preceq_R g_i^- \qquad \text{and} \qquad g_i^- \preceq_R g_i^+,$$

although

$$g_i^+ \neq g_i^-.$$

Thus $\preceq_R$ is naturally a preorder on X_∞, not a partial order.

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3 Support Equivalence

Definition 3.1 (Real-Support Equivalence)**.** Let

$$g, h \in X_\infty.$$

Define

$$\boxed{g \sim_R h \quad \Longleftrightarrow \quad R_g = R_h.}$$

The relation $\sim_R$ identifies epistemic cores with the same real coordinate support. Equivalently, the two cores represent the same set of real variables, even if their Boolean actions differ.

Proposition 3.2 (Real-Support Equivalence)**.** *The relation*

$$\sim_R$$

is an equivalence relation on X_∞*.*

Proof. Reflexivity follows from

$$R_g = R_g.$$

Symmetry follows because equality of sets is symmetric.
Transitivity follows because equality of sets is transitive.
Therefore $\sim_R$ is an equivalence relation. □

4 Support Classes

Definition 4.1 (Support-Class Space)**.** Define

$$\boxed{\mathcal{S}_R := X_\infty / \sim_R.}$$

For

$$g \in X_\infty,$$

write

$$[g]_R$$

for its equivalence class under $\sim_R$. Thus

$$\boxed{[g]_R = \{h \in X_\infty : R_h = R_g\}.}$$

The class $[g]_R$ is the support class of g.

Remark 4.2 (What a Support Class Retains). Passing from an epistemic core $g \in X_\infty$ to its class $[g]_R \in \mathcal{S}_R$ intentionally forgets the truth-table content of the core and retains only its real-support content. Distinct epistemic cores may therefore belong to the same support class.

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5 Order on the Quotient

Definition 5.1 (Order on Support Classes)**.** Let

$$[g]_R, [h]_R \in \mathcal{S}_R.$$

Define

$$\boxed{[g]_R \leq_R [h]_R \quad \Longleftrightarrow \quad R_g \subseteq R_h.}$$

Lemma 5.2 (Well-Definedness of $\leq_R$)**.** *The relation $\leq_R$ is well-defined on $\mathcal{S}_R$.*

Proof. Suppose that

$$g \sim_R g' \qquad \text{and} \qquad h \sim_R h'.$$

Then

$$R_g = R_{g'}$$

and

$$R_h = R_{h'}.$$

Therefore

$$R_g \subseteq R_h$$

if and only if

$$R_{g'} \subseteq R_{h'}.$$

Hence the definition of

$$[g]_R \leq_R [h]_R$$

does not depend on the chosen representatives. □

Theorem 5.3 (Partial Order on Support Classes)**.** *The relation*

$$\leq_R$$

is a partial order on

$$\mathcal{S}_R.$$

Proof. We prove reflexivity, transitivity, and antisymmetry.

First, for every

$$[g]_R \in \mathcal{S}_R,$$

we have

$$R_g \subseteq R_g.$$

Hence

$$[g]_R \leq_R [g]_R.$$

Thus $\leq_R$ is reflexive.

Second, suppose that

$$[g]_R \leq_R [h]_R \qquad \text{and} \qquad [h]_R \leq_R [k]_R.$$

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Then

$$R_g \subseteq R_h$$

and

$$R_h \subseteq R_k.$$

Therefore

$$R_g \subseteq R_k,$$

so

$$[g]_R \leq_R [k]_R.$$

Thus $\leq_R$ is transitive.

Finally, suppose that

$$[g]_R \leq_R [h]_R \qquad \text{and} \qquad [h]_R \leq_R [g]_R.$$

Then

$$R_g \subseteq R_h$$

and

$$R_h \subseteq R_g.$$

Hence

$$R_g = R_h.$$

Thus

$$g \sim_R h,$$

and therefore

$$[g]_R = [h]_R.$$

So $\leq_R$ is antisymmetric.

Hence $\leq_R$ is a partial order on $\mathcal{S}_R$. □

6 External Representation of Support Classes

Definition 6.1 (Support Representation Map)**.** Define

$$\Phi : \mathcal{S}_R \to [\mathbb{N}]^{<\omega}$$

by

$$\boxed{\Phi([g]_R) := R_g.}$$

Lemma 6.2 (Well-Definedness of Φ)**.** *The map*

$$\Phi([g]_R) = R_g$$

is well-defined.

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Proof. If

$$[g]_R = [h]_R,$$

then

$$g \sim_R h.$$

By definition of $\sim_R$,

$$R_g = R_h.$$

Therefore

$$\Phi([g]_R) = \Phi([h]_R).$$

□

Proposition 6.3 (External Representation)**.** *The map*

$$\Phi([g]_R) = R_g$$

is an order isomorphism between $(\mathcal{S}_R, \leq_R)$ *and its image in*

$$([\mathbb{N}]^{<\omega}, \subseteq).$$

In particular,

$$\boxed{[g]_R \leq_R [h]_R \quad \Longleftrightarrow \quad \Phi([g]_R) \subseteq \Phi([h]_R).}$$

Proof. By the definition of $\leq_R$,

$$[g]_R \leq_R [h]_R$$

if and only if

$$R_g \subseteq R_h.$$

By the definition of Φ, this is equivalent to

$$\Phi([g]_R) \subseteq \Phi([h]_R).$$

If

$$\Phi([g]_R) = \Phi([h]_R),$$

then

$$R_g = R_h.$$

Hence

$$g \sim_R h,$$

and therefore

$$[g]_R = [h]_R.$$

Thus Φ is injective. Consequently, Φ represents the quotient order as an inclusion order on its image. □

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Remark 6.4 (Exact Status of the Support Representation). The map

$$\Phi([g]_R) = R_g$$

is an exact order isomorphism between

$$(\mathcal{S}_R, \leq_R)$$

and its image

$$\Phi(\mathcal{S}_R) \subseteq [\mathbb{N}]^{<\omega}$$

ordered by inclusion.

Thus, at the level established in the present paper, no structural distinction is claimed between the support order and the inclusion order on its image. Whether the image is the whole of

$$[\mathbb{N}]^{<\omega}$$

is a separate realization question not settled by this proposition alone.

The terminology and genealogy assigned to this order in the present volume are separate from this exact order-theoretic identification.

7 Formal Statement of Variant 12

On the level of epistemic cores, the construction begins from the prequotient support relation

$$g \preceq_R h \quad \Longleftrightarrow \quad R_g \subseteq R_h.$$

On the level of support classes, Variant 12 takes its exact ordered form:

$$\boxed{\mathbf{V}_{12}: \quad [g]_R \leq_R [h]_R \Longleftrightarrow R_g \subseteq R_h.}$$

Thus:

$$\boxed{\text{Variant 12 establishes the order of real support.}}$$

The order compares support classes exclusively through their real-support content:

$$[g]_R \leq_R [h]_R \quad \Longleftrightarrow \quad R_g \subseteq R_h.$$

Accordingly, at the quotient level it is precisely the *order of real support*. The quotient does not retain the truth-table content of the represented epistemic cores.

8 Compatibility with Epistemic Width and Layers

Recall that epistemic width is given by

$$\boxed{\mathrm{PE}_B(g) = |\operatorname{Real}(g)| = |R_g|.}$$

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Proposition 8.1 (Monotonicity of Epistemic Width)**.** *If*

$$g \preceq_R h,$$

then

$$\boxed{\mathrm{PE}_B(g) \leq \mathrm{PE}_B(h).}$$

Proof. If

$$g \preceq_R h,$$

then

$$R_g \subseteq R_h.$$

Since the real supports are finite,

$$|R_g| \leq |R_h|.$$

By the definition of epistemic width,

$$\mathrm{PE}_B(g) = |\operatorname{Real}(g)| = |R_g|$$

and

$$\mathrm{PE}_B(h) = |\operatorname{Real}(h)| = |R_h|.$$

Therefore

$$\mathrm{PE}_B(g) \leq \mathrm{PE}_B(h).$$

□

The converse implication does not hold in general:

$$\boxed{\mathrm{PE}_B(g) \leq \mathrm{PE}_B(h) \not\Longrightarrow g \preceq_R h.}$$

Indeed, let

$$g : \{0,1\}^{\{1\}} \to \{0,1\}, \qquad g(u) = u_1,$$

and

$$h : \{0,1\}^{\{2\}} \to \{0,1\}, \qquad h(v) = v_2.$$

Then

$$R_g = \{1\}, \qquad R_h = \{2\},$$

so

$$\mathrm{PE}_B(g) = \mathrm{PE}_B(h) = 1.$$

However,

$$R_g \not\subseteq R_h,$$

and therefore

$$g \not\preceq_R h.$$

Width comparison records only cardinality, whereas $\preceq_R$ records inclusion of real supports.

Let

$$L_d = \{g \in X_\infty : \mathrm{PE}_B(g) = d\}$$

denote the width layers.

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Corollary 8.2 (Compatibility with Layers)**.** *If*

$$g \in L_d, \qquad h \in L_e, \qquad g \preceq_R h,$$

then

$$\boxed{d \leq e.}$$

Proof. Since

$$g \in L_d,$$

we have

$$\mathrm{PE}_B(g) = d.$$

Since

$$h \in L_e,$$

we have

$$\mathrm{PE}_B(h) = e.$$

By monotonicity of epistemic width,

$$\mathrm{PE}_B(g) \leq \mathrm{PE}_B(h).$$

Therefore

$$d \leq e.$$

□

Variant 12 orders real supports, not the layers themselves.

9 Conclusion

This paper formulated Variant 12 as the order of real support.

The support preorder

$$g \preceq_R h \quad \Longleftrightarrow \quad R_g \subseteq R_h$$

is a preorder on X_∞.

The equivalence relation

$$g \sim_R h \quad \Longleftrightarrow \quad R_g = R_h$$

identifies epistemic cores with the same real support.

The quotient

$$\mathcal{S}_R = X_\infty / \sim_R$$

carries the partial order

$$[g]_R \leq_R [h]_R \quad \Longleftrightarrow \quad R_g \subseteq R_h.$$

The map

$$\Phi([g]_R) = R_g$$

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identifies the quotient order with inclusion of coordinate supports. The information ordered by $(\leq_R)$ is exactly the real-support content retained by the support classes.

The support order is compatible with epistemic width:

$$g \preceq_R h \quad \Longrightarrow \quad \mathrm{PE}_B(g) \leq \mathrm{PE}_B(h),$$

but width comparison does not determine support inclusion in general.

It is also compatible with width layers:

$$g \in L_d, \quad h \in L_e, \quad g \preceq_R h \quad \Longrightarrow \quad d \leq e.$$

Thus:

Paper VIII establishes the order of real support.

It establishes order, not composition.

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Paper IX

Directional Asymmetry after Variant 12. Why the Theory Privileges Minimal Common Real Extension over Greatest Common Part

Piotr Dariusz Wójcik

Version 1.3

Abstract

Paper VIII established Variant 12 as the order of real support on the space of support classes

$$(\mathcal{S}_R, \leq_R).$$

Once this order has been introduced, two binary extremal structures become available. The first is the minimal common real extension

$$[g]_R \vee_R [h]_R,$$

externally represented by the union of coordinate supports. The second is the greatest common part

$$[g]_R \wedge_R [h]_R,$$

externally represented by their intersection.

At the level of the abstract partial order, these two structures are dual. Hence the order alone does not yet justify privileging one of them as the next structural act of the theory. The present paper resolves this difficulty by showing that the relevant asymmetry does not arise from the order considered in isolation, but from the internal logic of the theory preceding Variant 12.

Earlier stages of the theory established a principle of least sufficient real support: the full real determination of a function is preserved, while fictitious coordinate occurrence is removed. After Variant 12, the same preservation/leastness pattern becomes applicable to a new setting made available by the support order. It generates a question within the theory: what is the least support class that preserves two already established support classes simultaneously? The answer is precisely

$$[g]_R \vee_R [h]_R.$$

By contrast,

$$[g]_R \wedge_R [h]_R$$

answers a different question: it extracts the greatest common part of two supports and preserves both full support classes only in the degenerate case in which those classes are equal.

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Thus the theory possesses a genuine directional asymmetry after Variant 12. It privileges minimal common real extension over greatest common part not because join is mathematically superior to meet, but because join answers the question actually generated by the previous development of the theory. This result establishes the proper theoretical ground for the later formulation of Variant 13.

Keywords: Variant 12, real support, directional asymmetry, minimal common real extension, greatest common part, support classes, epistemic minimality, Variant 13.

1 Introduction

Paper VIII established Variant 12 as the order of real support. Its central structure is the partially ordered space

$$(\mathcal{S}_R, \leq_R),$$

where

$$\mathcal{S}_R = X_\infty/\!\sim_R$$

is the space of real-support classes. For every epistemic core $g \in X_\infty$, the set $\mathrm{Real}(g)$ records its real variables, while

$$R_g = \{i : x_i \in \mathrm{Real}(g)\}$$

is their coordinate representation. The order is defined by

$$[g]_R \leq_R [h]_R \quad \Longleftrightarrow \quad R_g \subseteq R_h.$$

This order is not an external decoration added to the theory. It is the first order structure arising after Variant 11 has separated syntactic occurrence from real determination and after Paper VII has established the space of finite epistemic cores.

Once Variant 12 is in place, two binary extremal constructions become available for support classes:

$$[g]_R \vee_R [h]_R,$$

the minimal common real extension, and

$$[g]_R \wedge_R [h]_R,$$

the greatest common part.

At the level of the abstract order

$$(\mathcal{S}_R, \leq_R),$$

these two constructions are dual. Therefore the order itself, considered in isolation, does not yet determine which one should guide the next act of the theory.

This creates the central question of the present paper:

Does the theory after Variant 12 possess an internal

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directional asymmetry that privileges minimal common

real extension over greatest common part?

The answer developed here is affirmative.

The paper does not claim that

$$\vee_R$$

is mathematically more important than

$$\wedge_R.$$

Such a claim would be false at the level of the abstract order. Instead, the paper shows that the earlier logic of the theory generates a question of *minimal common preservation of real support*, and that this question is answered by

$$\vee_R,$$

not by

$$\wedge_R.$$

The decisive point is therefore not that one extremal construction defeats the other. The decisive point is that:

the theory itself generates a question,

and only one of the two extremal structures

answers that question.

2 The Support Order after Variant 12

We briefly recall the structure established in Paper VIII.

Let

$$X_\infty$$

be the space of finite epistemic cores. For

$$g \in X_\infty,$$

the set of its real variables is denoted by

$$\mathrm{Real}(g),$$

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and its real coordinate support is

$$R_g = \{i : x_i \in \mathrm{Real}(g)\}.$$

Two epistemic cores are real-support equivalent when:

$$g \sim_R h \quad \Longleftrightarrow \quad R_g = R_h.$$

Equivalently, the two cores represent the same set of real variables.

The quotient space

$$\mathcal{S}_R = X_\infty / \sim_R$$

is the space of support classes. For

$$g \in X_\infty,$$

we write

$$[g]_R$$

for its equivalence class.

Variant 12 establishes the order:

$$[g]_R \leq_R [h]_R \quad \Longleftrightarrow \quad R_g \subseteq R_h.$$

Paper VIII also introduced the external support representation

$$\Phi : \mathcal{S}_R \to [\mathbb{N}]^{<\omega},$$

defined by

$$\Phi([g]_R) = R_g.$$

Then:

$$[g]_R \leq_R [h]_R \quad \Longleftrightarrow \quad \Phi([g]_R) \subseteq \Phi([h]_R).$$

Under the support representation (Φ), the order is exactly inclusion of finite coordinate supports. At the level of the quotient, it compares the real-support content retained by the support classes.

The direction

$$[g]_R \leq_R [h]_R$$

means that the coordinate support representing the real-variable content of $[g]_R$ is retained within $[h]_R$.

Variant 12 is the order of preservation and extension

of real-support content.

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3 Two Extremal Support Structures

Once the order

$$(\mathcal{S}_R, \leq_R)$$

has been established, two binary extremal structures appear naturally.

3.1 Minimal Common Real Extension

Definition 3.1 (Minimal common real extension)**.** Let

$$[g]_R, [h]_R \in \mathcal{S}_R.$$

A class

$$U \in \mathcal{S}_R$$

is a *minimal common real extension* of $[g]_R$ and $[h]_R$ if:

$$[g]_R \leq_R U,$$

$$[h]_R \leq_R U,$$

and for every

$$[k]_R \in \mathcal{S}_R,$$

$$([g]_R \leq_R [k]_R \ \wedge \ [h]_R \leq_R [k]_R) \Longrightarrow U \leq_R [k]_R.$$

Here "minimal" means least with respect to the real-support order among all common real extensions.

When such an object exists, it is unique by antisymmetry of $\leq_R$ and is denoted:

$$[g]_R \vee_R [h]_R.$$

Thus:

$$[g]_R \leq_R [g]_R \vee_R [h]_R,$$

$$[h]_R \leq_R [g]_R \vee_R [h]_R,$$

and:

$$([g]_R \leq_R [k]_R \ \wedge \ [h]_R \leq_R [k]_R) \Longrightarrow [g]_R \vee_R [h]_R \leq_R [k]_R.$$

3.2 Greatest Common Part

Definition 3.2 (Greatest common part)**.** Let

$$[g]_R, [h]_R \in \mathcal{S}_R.$$

A class

$$M \in \mathcal{S}_R$$

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is a *greatest common part* of $[g]_R$ and $[h]_R$ if:

$$M \leq_R [g]_R,$$

$$M \leq_R [h]_R,$$

and for every

$$[k]_R \in \mathcal{S}_R,$$

$$([k]_R \leq_R [g]_R \ \wedge \ [k]_R \leq_R [h]_R) \Longrightarrow [k]_R \leq_R M.$$

When such an object exists, it is unique by antisymmetry of $\leq_R$ and is denoted:

$$[g]_R \wedge_R [h]_R.$$

Thus:

$$[g]_R \wedge_R [h]_R \leq_R [g]_R,$$

$$[g]_R \wedge_R [h]_R \leq_R [h]_R,$$

and:

$$([k]_R \leq_R [g]_R \ \wedge \ [k]_R \leq_R [h]_R) \Longrightarrow [k]_R \leq_R [g]_R \wedge_R [h]_R.$$

Lemma 3.3 (Finite Support Realization)**.** *For every*

$$K \in [\mathbb{N}]^{<\omega},$$

there exists an epistemic core

$$e_K \in X_\infty$$

such that

$$R_{e_K} = K.$$

Corollary 3.4 (Full Finite-Support Representation)**.** *The support representation map*

$$\Phi : \mathcal{S}_R \to [\mathbb{N}]^{<\omega}$$

is surjective. Consequently,

$$\boxed{\Phi : (\mathcal{S}_R, \leq_R) \xrightarrow{\cong} ([\mathbb{N}]^{<\omega}, \subseteq)}$$

is an order isomorphism.

Proof. Let

$$K \in [\mathbb{N}]^{<\omega}.$$

By the Finite Support Realization Lemma, there exists

$$(e_K \in X_\infty) such that R_{e_K} = K.$$

Hence

$$\Phi([e_K]_R) = K.$$

Therefore (Φ) is surjective. Paper VIII already established that (Φ) is an order isomorphism onto its image, so the conclusion follows. □

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Proof. If $K \neq \varnothing$, define

$$e_K : \{0,1\}^K \to \{0,1\}$$

by

$$e_K(x) = \bigwedge_{i \in K} x_i.$$

For every $i \in K$, fixing all other coordinates at 1 shows that flipping the i-th coordinate changes the value of e_K. Hence every coordinate in K is real and

$$R_{e_K} = K.$$

If $K = \varnothing$, either constant epistemic core on the empty domain has empty coordinate support. Thus the assertion holds for every finite K. □

Proposition 3.5 (Existence and External Representation of Binary Extremal Classes)**.** *For every*

$$[g]_R, [h]_R \in \mathcal{S}_R,$$

both

$$[g]_R \vee_R [h]_R$$

and

$$[g]_R \wedge_R [h]_R$$

exist in $\mathcal{S}_R$*. Moreover,*

$$\Phi([g]_R \vee_R [h]_R) = R_g \cup R_h,$$

and

$$\Phi([g]_R \wedge_R [h]_R) = R_g \cap R_h.$$

Proof. Since

$$R_g \quad \text{and} \quad R_h$$

are finite, both

$$R_g \cup R_h$$

and

$$R_g \cap R_h$$

are finite. By the finite support realization lemma, there exist support classes whose coordinate supports are precisely

$$R_g \cup R_h$$

and

$$R_g \cap R_h.$$

The class represented by

$$R_g \cup R_h$$

is an upper bound of both $[g]_R$ and $[h]_R$, because

$$R_g \subseteq R_g \cup R_h \qquad \text{and} \qquad R_h \subseteq R_g \cup R_h.$$

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If $[k]_R$ is any other common upper bound, then

$$R_g \subseteq R_k \qquad \text{and} \qquad R_h \subseteq R_k,$$

so

$$R_g \cup R_h \subseteq R_k.$$

Hence this class is

$$[g]_R \vee_R [h]_R.$$

The class represented by

$$R_g \cap R_h$$

is a lower bound of both $[g]_R$ and $[h]_R$, because

$$R_g \cap R_h \subseteq R_g \qquad \text{and} \qquad R_g \cap R_h \subseteq R_h.$$

If $[k]_R$ is any other common lower bound, then

$$R_k \subseteq R_g \qquad \text{and} \qquad R_k \subseteq R_h,$$

so

$$R_k \subseteq R_g \cap R_h.$$

Hence this class is

$$[g]_R \wedge_R [h]_R.$$

The stated identities under Φ follow directly from the definition of the external support representation. □

Remark 3.6 (Exact Transport of the Extremal Operations). Under the full support isomorphism Φ,

$$\boxed{\Phi(A \vee_R B) = \Phi(A) \cup \Phi(B)}$$

and

$$\boxed{\Phi(A \wedge_R B) = \Phi(A) \cap \Phi(B).}$$

Thus the join and meet on support classes are exactly the operations of union and intersection transported through Φ.

Equivalently, the ordered structure together with its two binary extremal operations is identified exactly with

$$\left([\mathbb{N}]^{<\omega}, \subseteq, \cup, \cap\right).$$

No weaker notion of representation is intended here.

Remark 3.7 (Mathematical Status of the Extremal Operations). Since for every pair

$$A, B \in \mathcal{S}_R$$

the least upper bound $(A \vee_R B)$ and the greatest lower bound $(A \wedge_R B)$ exist uniquely, the assignments

$$\vee_R : \mathcal{S}_R \times \mathcal{S}_R \to \mathcal{S}_R$$

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and

$$\wedge_R : \mathcal{S}_R \times \mathcal{S}_R \to \mathcal{S}_R$$

are already well-defined binary operations at the level of the support order. The question remaining in this paper is not their mathematical existence, but whether the preceding development of the theory selects one of them as the proper direction of continuation.

3.3 The Apparent Symmetry

At the level of the abstract partial order, both constructions are legitimate. The first points upward in the order, while the second points downward.

Remark 3.8 (Order-Theoretic Symmetry). At the level of the abstract partial order, the extremal constructions $\vee_R$ and $\wedge_R$ are dual. Their order-theoretic existence alone does not privilege either of them as the next act of the theory.

4 Why the Order Alone Does Not Determine Direction

Suppose that

$$(\mathcal{S}_R, \leq_R)$$

is considered only as an abstract partially ordered set.

Then:

$$\vee_R$$

and

$$\wedge_R$$

are structurally dual extremal constructions.

No internal asymmetry arises merely from their existence.

Therefore, it would be illegitimate to claim:

$$\boxed{\vee_R \text{ is the next act of the theory simply because it exists.}}$$

The same reasoning would equally support:

$$\wedge_R.$$

The problem must therefore be moved to a deeper level.

Variant 12 did not introduce an arbitrary order. It introduced an order with internal theoretical meaning:

$$[g]_R \leq_R [h]_R$$

means that the real support represented by $[g]_R$ is retained within the real support represented by $[h]_R$.

Thus the order has a direction of increasing real-support inclusion:

$$[g]_R \quad \longrightarrow \quad [h]_R.$$

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This arrow is only a schematic representation of the direction $[g]_R \leq_R [h]_R$; it is not a procedure, trajectory, or dynamical transition.

Yet this alone is still insufficient. We must ask whether this direction is merely notational, or whether it continues a structural principle already present in the earlier development of the theory.

If an asymmetry exists, it must come from

the earlier logic of the theory,

not from the abstract order by itself.

5 The Earlier Logic of the Theory

The theory did not begin with the order of supports. It began with a distinction.

Variant 11 established the separation between:

syntactic occurrence

and

real determination.

A syntactically present variable need not participate in real determination. This separation led to canonical reduction from a syntactic function to its epistemic core.

For a syntactic function

$$f : \{0,1\}^N \to \{0,1\},$$

its real coordinate support is:

$$R_f = \{i : x_i \in \mathrm{Real}(f)\}.$$

Its epistemic core is:

$$f_\infty = f_{R_f}.$$

The canonical reduction:

$$f = f_\infty \circ \pi^N_{R_f}$$

removes fictitious coordinate occurrence while preserving the full real dependence of the original function.

The important point is not merely that something is removed. The important point is *what* is removed and *what* is preserved.

The theory removes fictitious excess,

but never discards real determination.

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This establishes a structural principle that precedes Variant 12:

$$\boxed{\text{preserve all real determination,}}$$

$$\boxed{\text{eliminate fictitious coordinate occurrence.}}$$

This is the source from which any later asymmetry must arise.

6 Minimal Sufficient Real Support for a Single Object

The earlier minimal-support theorem states that for a function

$$f : \{0,1\}^N \to \{0,1\},$$

the coordinate support

$$R_f$$

is the least set of coordinates sufficient to determine f.

More precisely, if

$$S \subseteq N$$

is such that f depends only on S, then:

$$R_f \subseteq S.$$

Hence:

$$\boxed{R_f \text{ is the least coordinate support sufficient to determine } f.}$$

Equivalently, $\mathrm{Real}(f)$ contains exactly the real variables corresponding to this least coordinate support.

This is a decisive fact.

The theory does not minimize by destroying real content. It minimizes by removing everything not necessary for preserving that content.

$$\boxed{\text{Minimality in the theory is not minimality by loss.}}$$

$$\boxed{\text{It is minimality under full preservation of real determination.}}$$

Thus the governing pattern before Variant 12 is:

$$\boxed{\text{retain everything real;}}$$

$$\boxed{\text{remove fictitious coordinate occurrence.}}$$

This pattern is already fully established before the question of $\vee_R$ and $\wedge_R$ arises.

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7 Does Variant 12 Generate a Problem of Common Preservation?

We now reach the central threshold.

After Variant 12, the theory no longer contains only individual epistemic cores and their supports. It contains a whole ordered space of support classes:

$$(\mathcal{S}_R, \leq_R).$$

Let:

$$[g]_R, [h]_R \in \mathcal{S}_R.$$

Once two independently established support classes are considered together within the same ordered space, the theory can ask for a class preserving both of them without reducing either to the other. This co-presence generates the question of their possible common preservation.

The order can compare them:

$$[g]_R \leq_R [h]_R,$$

or:

$$[h]_R \leq_R [g]_R,$$

or determine that they are incomparable:

$$[g]_R \not\leq_R [h]_R, \qquad [h]_R \not\leq_R [g]_R.$$

However, once two support classes are given together in the same ordered space, a new question is generated by the internal logic of the theory:

$$\boxed{\text{Is there a support class that preserves}}$$

$$\boxed{\text{the full real-support content of both classes at once?}}$$

For $A, U \in \mathcal{S}_R$, we say that U preserves A when

$$A \leq_R U.$$

Formally, we now ask whether there exists

$$U \in \mathcal{S}_R$$

such that:

$$[g]_R \leq_R U,$$
$$[h]_R \leq_R U.$$

The first relation means that U preserves the real-support content of $[g]_R$. The second means that U preserves the real-support content of $[h]_R$. Together:

$$[g]_R \leq_R U, \qquad [h]_R \leq_R U$$

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express:

$$\boxed{U \text{ preserves both real-support structures simultaneously.}}$$

This question is not imported from external order theory. It arises from the internal meaning of Variant 12 once more than one support class is considered. Order theory supplies the formal shape of the answer; the internal development of the theory supplies the question to be answered.

$$\boxed{\text{After Variant 12, the theory genuinely generates}}$$

$$\boxed{\text{the problem of common preservation of two real supports.}}$$

8 From Common Preservation to Minimal Common Preservation

Once the theory asks for a class U preserving both support classes, the earlier preservation/leastness pattern becomes applicable to this newly generated problem.

The theory cannot be satisfied with an arbitrary common extension.

Suppose:

$$[g]_R \leq_R [k]_R,$$
$$[h]_R \leq_R [k]_R.$$

Then $[k]_R$ preserves both real supports. But it may contain real support beyond what is necessary for this simultaneous preservation. Such additional support need not be fictitious for an epistemic core representing $[k]_R$; it is unnecessary only relative to the present task of preserving the two given classes.

The internal preservation/leastness pattern therefore selects the least such class.

Thus we seek $U \in \mathcal{S}_R$ satisfying:

$$[g]_R \leq_R U,$$
$$[h]_R \leq_R U,$$

and:

$$([g]_R \leq_R [k]_R \ \wedge \ [h]_R \leq_R [k]_R) \Longrightarrow U \leq_R [k]_R.$$

This is exactly the defining property of:

$$[g]_R \vee_R [h]_R.$$

Hence:

$$\boxed{[g]_R \vee_R [h]_R}$$

is:

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$$\boxed{\text{the least real-support class sufficient}}$$

$$\boxed{\text{to preserve both support classes simultaneously.}}$$

The structural analogy with the earlier minimal-support theorem is now clear.

For one object:

$$R_f$$

is the least coordinate support sufficient to preserve its full real determination.

For two support classes:

$$[g]_R \vee_R [h]_R$$

is the least support class sufficient to preserve both real supports together.

The analogy concerns leastness under preservation. It does not identify canonical reduction of a function with common extension of two support classes.

9 Preservation of Both Supports and Elimination of Excess

The minimal common real extension

$$[g]_R \vee_R [h]_R$$

realizes the two principles that governed the earlier theory.

Preservation

We have:

$$[g]_R \leq_R [g]_R \vee_R [h]_R,$$

$$[h]_R \leq_R [g]_R \vee_R [h]_R.$$

Therefore:

$$\boxed{[g]_R \vee_R [h]_R \text{ preserves the full real-support content}}$$

$$\boxed{\text{of both } [g]_R \text{ and } [h]_R.}$$

Elimination of excess

If $[k]_R$ is any other class satisfying:

$$[g]_R \leq_R [k]_R,$$

$$[h]_R \leq_R [k]_R,$$

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then:

$$[g]_R \vee_R [h]_R \leq_R [k]_R.$$

Thus:

$$\boxed{[g]_R \vee_R [h]_R \text{ contains no support beyond what is necessary}}$$

$$\boxed{\text{for preserving both given real-support classes.}}$$

Remark 9.1 (Two Kinds of Excess). At the level of a single syntactic function, canonical reduction removes fictitious coordinate occurrence. At the level of two support classes, minimal common real extension excludes support that is unnecessary relative to the task of preserving those two classes. The second kind of excess need not be fictitious for an epistemic core representing a larger support class.

The principle is therefore an analogous preservation/leastness pattern:

$$\boxed{\text{preserve all real necessity;}}$$

$$\boxed{\text{remove every unnecessary excess relative to the task.}}$$

10 The Status of $\vee_R$

The construction

$$\vee_R$$

is not privileged because it is a join in an abstract poset.

It is privileged because it answers the specific question generated by the internal development of the theory:

$$\boxed{\text{What is the minimal support class}}$$

$$\boxed{\text{that preserves two real supports together?}}$$

The answer is:

$$\boxed{[g]_R \vee_R [h]_R.}$$

Externally:

$$\Phi([g]_R \vee_R [h]_R) = R_g \cup R_h.$$

Internally:

$$\boxed{[g]_R \vee_R [h]_R \text{ is the minimal common real extension}}$$

$$\boxed{\text{of two support classes.}}$$

Under the support isomorphism Φ, the operation $\vee_R$ is exactly union transported to the support-class space:

$$\Phi(A \vee_R B) = \Phi(A) \cup \Phi(B).$$

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The distinction retained by the present paper is therefore not a structural distinction between $\vee_R$ and union. It concerns the theoretical question that selects this join, rather than the dual meet, as the direction of continuation: minimal common preservation of two support classes.

Externally: union of coordinate supports.

Internally: minimal common preservation

of real-support content.

11 The Status of $\wedge_R$

The construction

$$\wedge_R$$

is also genuine and meaningful.

It answers the question:

What is the greatest support class

contained in both given support classes?

Externally:

$$\Phi([g]_R \wedge_R [h]_R) = R_g \cap R_h.$$

Internally:

$$[g]_R \wedge_R [h]_R$$

extracts their greatest common part.

Unless the two support classes are equal, the greatest common part cannot preserve both of them in full. More precisely, the central theorem below proves that

$$[g]_R \wedge_R [h]_R \text{ preserves both full support classes} \quad \Longleftrightarrow \quad [g]_R = [h]_R.$$

It therefore answers a different question from the one generated by the minimal-preservation logic of the theory.

$\vee_R$ answers the question of minimal common preservation.

$\wedge_R$ answers the question of greatest common residue.

The second question is legitimate. It may become important later. But it is not the question whose necessity emerges immediately after Variant 12.

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12 Directional Asymmetry after Variant 12

We can now state the central result.

Theorem 12.1 (Directional Asymmetry under Common Preservation)**.** *Let*

$$A = [g]_R, \qquad B = [h]_R$$

be support classes in $\mathcal{S}_R$*.*

Then $A \vee_R B$ *preserves both support classes:*

$$A \leq_R A \vee_R B, \qquad B \leq_R A \vee_R B.$$

By contrast, $A \wedge_R B$ *preserves both support classes if and only if*

$$A = B.$$

Consequently, among the two binary extremal classes considered here, only

$$A \vee_R B$$

solves the problem of common preservation in general. Moreover, it is the least class that does so.

Proof. By the defining property of the minimal common real extension,

$$A \leq_R A \vee_R B \qquad \text{and} \qquad B \leq_R A \vee_R B.$$

Hence $A \vee_R B$ preserves the full real-support content of both classes.

For the greatest common part, one always has

$$A \wedge_R B \leq_R A \qquad \text{and} \qquad A \wedge_R B \leq_R B.$$

Suppose that $A \wedge_R B$ also preserves both classes. Then

$$A \leq_R A \wedge_R B \qquad \text{and} \qquad B \leq_R A \wedge_R B.$$

By antisymmetry,

$$A = A \wedge_R B \qquad \text{and} \qquad B = A \wedge_R B,$$

so

$$A = B.$$

Conversely, if $A = B$, then

$$A \wedge_R B = A = B,$$

and the greatest common part preserves both classes.

Thus $A \wedge_R B$ preserves both support classes if and only if $A = B$. Therefore, among the two extremal constructions considered here, only $A \vee_R B$ solves the problem of common preservation for arbitrary support classes. Its leastness follows from the defining property of the minimal common real extension. □

Remark 12.2 (Theoretical Direction). The theorem does not assert that $\vee_R$ is mathematically superior to $\wedge_R$. It shows that only $\vee_R$ answers the problem of minimal common preservation generated by the preceding development of the theory. This is the directional ground for the later formulation of Variant 13.

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13 The Route toward Variant 13

The existence of the dual extremal structure

$$[g]_R \wedge_R [h]_R$$

shows that the support order alone does not determine the next theoretical act. The directional asymmetry established above resolves this ambiguity.

At the level of abstract order, both extremal constructions remain legitimate. At the level of theoretical continuation, only the minimal common real extension answers the problem of common preservation.

The internal logic of the theory does not ask first for the greatest common part. It asks for the minimal support structure that preserves two real supports together.

Therefore:

$$\boxed{\text{the route toward Variant 13 is now determined,}}$$

but for a deeper reason.

$$\boxed{\text{Variant 13 should be approached as the act of}}$$

$$\boxed{\text{minimal common real extension,}}$$

$$\boxed{\text{not as an arbitrary adoption of a join operation.}}$$

This paper does not yet designate either extremal operation as Variant 13. It establishes the directional asymmetry that selects the already defined join operation ($\vee_R$) as the theoretically justified continuation.

14 Conclusion

The order of real support

$$(\mathcal{S}_R, \leq_R)$$

admits two dual binary extremal structures:

$$[g]_R \vee_R [h]_R,$$

the minimal common real extension, and

$$[g]_R \wedge_R [h]_R,$$

the greatest common part.

At the level of the abstract order, neither of these structures is privileged.

However, the theory does not arise from the abstract order alone. For a single syntactic function, its earlier development established canonical reduction: all real determination is retained and fictitious coordinate occurrence is removed. In the later two-class problem,

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the corresponding preservation/leastness pattern seeks the least common support class and therefore excludes support unnecessary relative to that task.

retain everything real;

exclude only the excess relative to the task at hand.

After Variant 12, this pattern generates a new question:

What is the minimal real-support class

that preserves two real-support classes together?

The answer is:

$$[g]_R \vee_R [h]_R.$$

The construction

$$\wedge_R$$

is meaningful, but it answers another question. It extracts the greatest common part and preserves both full support classes only when those classes are equal.

The final conclusion is therefore:

The theory after Variant 12 possesses

an internal directional asymmetry.

This asymmetry privileges minimal common real extension

over greatest common part

as the proper direction toward Variant 13.

Thus Paper IX leaves the two extremal operations mathematically established, but selects $(\vee_R)$ as the theoretically justified direction toward Variant 13. The passage to Variant 13 will therefore be a designation and structural development of this already established join, not the first mathematical construction of the join itself.

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Paper X

Variant 13: Minimal Common Real Extension

Piotr Dariusz Wójcik

Version 1.3

Abstract

Paper VIII established Variant 12 as the order of real support on the space of support classes

$$(\mathcal{S}_R, \leq_R).$$

Paper IX then established that every pair of support classes possesses a unique minimal common real extension, denoted by

$$A \vee_R B,$$

and showed why this construction, rather than greatest common part, provides the proper theoretical continuation of Variant 12.

The present paper begins from this settled point. Paper IX has already established the well-defined join operation

$$\vee_R : \mathcal{S}_R \times \mathcal{S}_R \longrightarrow \mathcal{S}_R, \qquad (A, B) \longmapsto A \vee_R B,$$

where $(A \vee_R B)$ is the unique minimal common real extension of (A) and (B). The present paper designates this selected operation as Variant 13 and develops its first structural consequences. Externally,

$$[g]_R \vee_R [h]_R$$

is represented by the union of coordinate supports:

$$\Phi([g]_R \vee_R [h]_R) = R_g \cup R_h,$$

where

$$R_g = \{i : x_i \in \mathrm{Real}(g)\}.$$

Under the support isomorphism Φ, Variant 13 is exactly the union operation transported to the support-class space:

$$\Phi(A \vee_R B) = \Phi(A) \cup \Phi(B).$$

Within the theoretical development of this volume, the same operation is characterized as minimal common real extension: the least support class preserving the two given support classes simultaneously.

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The paper proves the basic structural properties of $\vee_R$: idempotence, commutativity, and associativity. Finally, it shows that the order of Variant 12 is characterized by absorption into the minimal common real extension:

$$[g]_R \leq_R [h]_R \quad \Longleftrightarrow \quad [g]_R \vee_R [h]_R = [h]_R.$$

Thus, within the dependence-and-support line developed from Variant 11 onward, Variant 13 is established as the next structural act:

$$\text{separation} \longrightarrow \text{order} \longrightarrow \text{minimal common real extension}.$$

Keywords: Variant 13, minimal common real extension, real support, support classes, epistemic cores, support order, binary extension, absorption characterization.

1 Introduction

Variant 11 established the separation between syntactic occurrence and real determination. Paper VII then placed epistemic cores in the global space

$$X_\infty.$$

Paper VIII introduced Variant 12 as the order of real support on the quotient space of support classes

$$(\mathcal{S}_R, \leq_R).$$

Finally, Paper IX established the unique minimal common real extension for every pair of support classes and showed that this construction, rather than greatest common part, provides the internally justified continuation of the support order.

Paper X therefore begins neither by reopening the directional problem nor by reopening the existence and uniqueness result already established in Paper IX. It begins from the join operation already determined there and designates that selected operation as the next structural act of the theory.

The present paper establishes:

$$\boxed{\text{Variant 13: Minimal Common Real Extension.}}$$

The guiding question inherited from Paper IX is:

$$\boxed{\text{Given two real-support classes, what is the least support class}}$$

$$\boxed{\text{that preserves both of them simultaneously?}}$$

Paper IX denoted the unique answer by:

$$[g]_R \vee_R [h]_R.$$

The purpose of the present paper is to recall the join operation established in Paper IX, designate it as Variant 13, and derive its first structural consequences.

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The Variant 11–13 line developed so far can therefore be read as:

$$\boxed{\text{Variant 11: separation;}}$$

$$\boxed{\text{Variant 12: order;}}$$

$$\boxed{\text{Variant 13: minimal common real extension.}}$$

2 The Point of Departure after Variant 12 and Paper IX

We recall the structure established before Variant 13.

Let

$$X_\infty$$

be the space of finite epistemic cores. For

$$g \in X_\infty,$$

the set of its real variables is denoted by

$$\mathrm{Real}(g),$$

while its real coordinate support is

$$R_g = \{i : x_i \in \mathrm{Real}(g)\}.$$

Two cores are real-support equivalent when:

$$g \sim_R h \quad \Longleftrightarrow \quad R_g = R_h.$$

Equivalently, the two cores represent the same set of real variables.

The quotient space

$$\mathcal{S}_R = X_\infty / \sim_R$$

is the space of real-support classes. The class of $g \in X_\infty$ is written:

$$[g]_R.$$

Variant 12 established the partial order:

$$[g]_R \leq_R [h]_R \quad \Longleftrightarrow \quad R_g \subseteq R_h.$$

Paper VIII also introduced the external support representation:

$$\Phi : \mathcal{S}_R \longrightarrow [\mathbb{N}]^{<\omega},$$

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defined by:

$$\Phi([g]_R) = R_g.$$

Thus:

$$[g]_R \leq_R [h]_R \quad \Longleftrightarrow \quad \Phi([g]_R) \subseteq \Phi([h]_R).$$

The same representation theorem established that Φ is injective on $\mathcal{S}_R$.

Paper IX established that every pair

$$A, B \in \mathcal{S}_R$$

possesses a unique minimal common real extension, denoted by

$$A \vee_R B.$$

It also resolved the directional question in favor of this construction. Paper X may therefore proceed directly from the already established join operation to its designation and structural development as Variant 13.

3 The Minimal Common Real Extension Recalled

We recall the minimal common real extension established in Paper IX, whose associated join operation is designated and developed here as Variant 13.

Definition 3.1 (Minimal Common Real Extension, recalled)**.** Let

$$A, B \in \mathcal{S}_R.$$

A class

$$U \in \mathcal{S}_R$$

is a minimal common real extension of A and B if:

$$A \leq_R U, \qquad B \leq_R U,$$

and for every

$$V \in \mathcal{S}_R,$$

the implication

$$(A \leq_R V \ \wedge \ B \leq_R V) \Longrightarrow U \leq_R V$$

holds.

Minimality is taken with respect to the order of Variant 12. The word "minimal" is used here in the strong order-theoretic sense of being least among all common real-support extensions.

Following Paper IX, for $A, U \in \mathcal{S}_R$, we say that U preserves A when

$$A \leq_R U.$$

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Thus U preserves both A and B when

$$A \leq_R U, \qquad B \leq_R U.$$

The intended theoretical meaning is:

$$\boxed{U \text{ is the least support class preserving } A \text{ and } B}$$

$$\boxed{\text{simultaneously.}}$$

Paper IX established that such a class exists uniquely for every pair $(A, B \in \mathcal{S}_R)$. Consequently, Paper IX already determines the well-defined binary operation

$$\vee_R : \mathcal{S}_R \times \mathcal{S}_R \to \mathcal{S}_R.$$

The present paper begins from this established operation.

4 The Binary Operation Established in Paper IX

Proposition 4.1 (Binary Operation Determined by Paper IX)**.** *For every*

$$A, B \in \mathcal{S}_R,$$

there exists exactly one minimal common real extension of A and B. Consequently, the assignment

$$(A, B) \longmapsto A \vee_R B$$

is well-defined on

$$\mathcal{S}_R \times \mathcal{S}_R.$$

Proof. Paper IX established existence and uniqueness of the minimal common real extension for every pair $(A, B \in \mathcal{S}_R)$. Existence supplies a value for every pair and uniqueness makes that value single-valued. Therefore

$$\vee_R : \mathcal{S}_R \times \mathcal{S}_R \to \mathcal{S}_R, \qquad (A, B) \longmapsto A \vee_R B,$$

is already a well-defined binary operation. □

This proposition records explicitly the operational status already implied by the results of Paper IX.

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5 Variant 13

Definition 5.1 (Variant 13: Minimal Common Real Extension)**.** The binary join operation established in Paper IX,

$$\boxed{\vee_R : \mathcal{S}_R \times \mathcal{S}_R \longrightarrow \mathcal{S}_R,}$$

is designated as *Variant 13*.

For

$$A, B \in \mathcal{S}_R,$$

its value

$$A \vee_R B$$

is the unique minimal common real extension of (A) and (B).

Variant 13 is not merely a binary operation written on support classes. Its theoretical content is the following:

$$\boxed{\text{two already established real-support classes}}$$

$$\boxed{\text{receive their least common real extension.}}$$

The operation $\vee_R$ is the formal carrier of this act.

Thus Variant 13 should be read internally as:

$$\boxed{\text{minimal common preservation of two real-support classes}}$$

$$\boxed{\text{without support excess relative to that preservation task.}}$$

Remark 5.2 (Support excess in Variant 13). The excess excluded by minimal common real extension is support not required for preserving the two given classes simultaneously. Such support need not be fictitious for an epistemic core representing a larger support class; it is unnecessary only relative to the present common-preservation task.

6 External Representation by Union of Coordinate Supports

The internal act of Variant 13 has a precise external representation.

Proposition 6.1 (External Support Representation of Variant 13)**.** *For every*

$$[g]_R, [h]_R \in \mathcal{S}_R,$$

we have:

$$\Phi([g]_R \vee_R [h]_R) = \Phi([g]_R) \cup \Phi([h]_R).$$

Since (Φ) is an isomorphism onto

$$[\mathbb{N}]^{<\omega},$$

this is an exact structural identification: $(\vee_R)$ is the union operation transported through (Φ).

Equivalently:

$$\boxed{\Phi([g]_R \vee_R [h]_R) = R_g \cup R_h.}$$

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Proof. Paper IX established that the unique minimal common real extension of $[g]_R$ and $[h]_R$ is represented by the coordinate support

$$R_g \cup R_h.$$

Variant 13 assigns precisely this unique class to the pair. Therefore

$$\Phi([g]_R \vee_R [h]_R) = R_g \cup R_h.$$

Since

$$\Phi([g]_R) = R_g, \qquad \Phi([h]_R) = R_h,$$

the result follows. □

This yields the essential double description of Variant 13:

Externally: union of coordinate supports.

Internally: minimal common real extension.

Thus there is no structural discrepancy between $(\vee_R)$ and union under the support isomorphism. The distinct statement made by Variant 13 concerns the role assigned to this already identified join within the development of the theory: it is the operation selected by the problem of minimal common preservation.

7 Basic Structural Properties of $\vee_R$

The operation $\vee_R$ possesses three basic structural properties: idempotence, commutativity, and associativity. These properties do not constitute a separate algebraic development. They express the immediate stability of the binary act established by Variant 13.

Under the support isomorphism (Φ), the operation $(\vee_R)$ is exactly transported to set union. The following identities may therefore be proved through the corresponding identities for union and transported back through the injective map (Φ). By Paper VIII, Φ is injective on $\mathcal{S}_R$. Therefore equality of external support representations implies equality of support classes.

Proposition 7.1 (Basic Structural Properties of $\vee_R$)**.** *For all*

$$A, B, C \in \mathcal{S}_R,$$

the following identities hold:

1. $A \vee_R A = A$;
2. $A \vee_R B = B \vee_R A$;
3. $(A \vee_R B) \vee_R C = A \vee_R (B \vee_R C)$.

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Proof. Let

$$\Phi(A) = G, \qquad \Phi(B) = H, \qquad \Phi(C) = K.$$

By the external support representation of Variant 13,

$$\Phi(A \vee_R B) = G \cup H.$$

For idempotence,

$$\Phi(A \vee_R A) = G \cup G = G = \Phi(A).$$

By injectivity of Φ,

$$A \vee_R A = A.$$

For commutativity,

$$\Phi(A \vee_R B) = G \cup H = H \cup G = \Phi(B \vee_R A).$$

By injectivity of Φ,

$$A \vee_R B = B \vee_R A.$$

For associativity,

$$\Phi((A \vee_R B) \vee_R C) = (G \cup H) \cup K,$$

while

$$\Phi(A \vee_R (B \vee_R C)) = G \cup (H \cup K).$$

Since union is associative,

$$(G \cup H) \cup K = G \cup (H \cup K).$$

By injectivity of Φ,

$$(A \vee_R B) \vee_R C = A \vee_R (B \vee_R C).$$

□

Internally, idempotence means that extending a support class with itself creates no new real support. Commutativity means that the order of two support classes does not affect their minimal common real extension. Associativity means that the minimal common real extension of finitely many classes does not depend on grouping. These are stability properties of common extension, not the beginning of a separate general algebra.

8 Order Characterization through $\vee_R$

Variant 13 does not merely add a new structure beside the order of Variant 12. It gives that order a canonical binary form.

Theorem 8.1 (Order Characterization by $\vee_R$)**.** *For all*

$$A, B \in \mathcal{S}_R,$$

the following equivalence holds:

$$\boxed{A \leq_R B \quad \Longleftrightarrow \quad A \vee_R B = B.}$$

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Proof. Assume first that

$$A \leq_R B.$$

Then B is a common real-support extension of A and B. Moreover, if U is any common extension of these two classes, then

$$B \leq_R U,$$

because U preserves B. Hence B is the minimal common real extension of A and B. Therefore

$$A \vee_R B = B.$$

Conversely, assume

$$A \vee_R B = B.$$

By the defining property of minimal common real extension,

$$A \leq_R A \vee_R B.$$

Substituting $A \vee_R B = B$, we obtain

$$A \leq_R B.$$

Thus

$$A \leq_R B \quad \Longleftrightarrow \quad A \vee_R B = B.$$

□

This theorem shows that the order of Variant 12 is characterized by absorption into the minimal common real extension of Variant 13.

Variant 13 does not stand beside Variant 12;

it gives the order of Variant 12 a canonical binary form.

9 Brief Remark on Finite Iteration

Associativity allows finite iteration of minimal common real extension.

Remark 9.1 (Finite iteration). For any finite family

$$[g_1]_R, \ldots, [g_n]_R \in \mathcal{S}_R, \qquad n \geq 1,$$

the expression

$$\bigvee_{i=1}^{n} [g_i]_R$$

is well-defined.

Its external support representation is:

$$\Phi\left(\bigvee_{i=1}^{n} [g_i]_R\right) = \bigcup_{i=1}^{n} R_{g_i}.$$

This remark records finite iteration only; no separate n-ary theory is developed here.

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10 Scope of the Paper

This paper establishes Variant 13 by designating the join operation

$$\vee_R : \mathcal{S}_R \times \mathcal{S}_R \to \mathcal{S}_R$$

already established in Paper IX as the selected binary act of minimal common real extension.

It develops this operation as Variant 13, proves its first structural properties, and characterizes the order of Variant 12 through it.

The present paper stops at Variant 13 as a binary operation on support classes and at its immediate structural consequences. No broader algebraic development is undertaken here.

11 Conclusion

Variant 13 has now been established as the designated binary act of minimal common real extension.

It is the join operation already mathematically established in Paper IX:

$$\boxed{\vee_R : \mathcal{S}_R \times \mathcal{S}_R \longrightarrow \mathcal{S}_R, \qquad (A, B) \longmapsto A \vee_R B.}$$

Internally, $A \vee_R B$ is the minimal common real extension of two support classes: the least support class that preserves both real-support contents simultaneously without support excess relative to that common-preservation task. At the variable level, this means preserving the real-variable content represented by both classes.

Externally:

$$\Phi([g]_R \vee_R [h]_R) = R_g \cup R_h.$$

The operation $\vee_R$ is:

$$\boxed{\text{idempotent,}}$$

$$\boxed{\text{commutative,}}$$

$$\boxed{\text{associative.}}$$

Moreover, the order of Variant 12 is characterized by:

$$\boxed{A \leq_R B \quad \Longleftrightarrow \quad A \vee_R B = B.}$$

Thus Variant 13 develops the next structural act within the dependence-and-support line begun with Variant 11:

$$\boxed{\text{separation} \longrightarrow \text{order} \longrightarrow \text{minimal common real extension.}}$$

The theory now designates as Variant 13 the join operation already established mathematically in Paper IX, selected there by the directional asymmetry after Variant 12 and developed here through its immediate structural laws.

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